

Placement Bridge

**A PROJECT REPORT
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**Under the Supervision of
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CERTIFICATE

Certified that **Devanshi Singhal 202410116100060, Deepanshu Ruhela 202410116100056, Bishop Tyagi 202410116100050, Dhvani Panchal 202410116100063** has/ have carried out the project work having “**Title of Report: Placement Bridge**” (CA401P) for **Master of Computer Applications** from KIET Group of Institutions (an Autonomous college) under Dr. A.P.J. Abdul Kalam Technical University (AKTU), Lucknow under my supervision. The project report embodies original work, and studies are carried out by the student himself/herself and the contents of the project report do not form the basis for the award of any other degree to the candidate or to anybody else from this or any other University/Institution.

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Placement Bridge
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Problem Statement

In today's competitive academic environment, placement preparation has become a major challenge for students due to the lack of structured guidance and reliable resources. While numerous online platforms provide interview preparation material, most of the content is generic, unorganized, and not aligned with institution-specific recruitment patterns. As a result, students often struggle to identify the right preparation strategy, leading to confusion, inefficient learning, and increased stress during placement season.

Another significant issue is the absence of a centralized platform that facilitates effective communication between senior and junior students. Senior students possess valuable insights into real interview experiences, company expectations, and preparation strategies, but this knowledge is often shared informally and remains inaccessible to a wider student community.

Due to this gap, juniors rely heavily on scattered sources such as social media, blogs, and external platforms, which may not always be authentic or relevant. This lack of structured mentorship and organized resources negatively impacts students' confidence and placement readiness.

Therefore, there is a need for a dedicated platform that connects seniors and juniors, enabling systematic knowledge sharing, mentorship, and access to authentic placement preparation resources in a centralized and user-friendly manner.

Keywords: Full Stack Development, Accessibility, Personalized Learning, Technology Integration.

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CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

In the modern educational environment, placement preparation has become one of the most significant phases in a student's academic journey. Students are expected to possess not only strong technical knowledge but also problem-solving abilities, communication skills, aptitude preparation, and awareness of company-specific recruitment processes. However, many students face difficulties in identifying the correct preparation strategy due to the absence of structured guidance and authentic resources.

Senior students who have successfully cleared placement processes hold valuable practical knowledge regarding interview experiences, preparation techniques, coding rounds, HR interviews, and company expectations. Unfortunately, this information is often shared informally through chats, social media, or personal interactions, making it inaccessible to a large number of students. As a result, juniors spend significant time searching for scattered resources and often remain confused about how to prepare effectively.

The **Placement Bridge** – Senior–Junior Connection Platform is developed as a centralized web-based platform that connects seniors and juniors within the academic ecosystem. The platform allows seniors to share placement experiences, interview questions, preparation roadmaps, and guidance videos, while juniors can access these resources in an organized and reliable manner. The system promotes mentorship, collaborative learning, and structured placement preparation.

The platform is designed using modern web technologies such as HTML, CSS, JavaScript, Node.js, Express.js, and MongoDB/MySQL to ensure efficient performance, scalability, and user-friendly interaction. Through this platform, students gain access to authentic placement guidance directly from seniors who have already experienced the recruitment process.

The overall objective of the system is to reduce confusion, improve placement readiness, and create a supportive learning environment that strengthens peer-to-peer mentorship within the institution.

1.2 PROBLEM STATEMENT

Placement preparation is often challenging for students because there is no centralized platform that provides structured guidance and institution-specific placement experiences. Although numerous online resources are available, most of them are generic, unorganized, or unrelated to the actual recruitment patterns faced by students in a particular college environment.

Junior students frequently encounter problems such as:

- Lack of proper guidance regarding placement preparation
- Uncertainty about which technologies or subjects to focus on
- Absence of company-specific interview experiences
- Limited access to authentic preparation strategies
- Dependence on scattered online resources and social media
- Stress and confusion during placement season

Another major issue is the communication gap between seniors and juniors. Seniors possess valuable real-world interview experiences, but there is no proper digital medium through which they can systematically share their knowledge with juniors. As a result, useful information remains restricted to a small group of students instead of benefiting the entire academic community.

Existing platforms such as YouTube, blogs, and online forums provide generalized content but fail to offer institution-specific placement guidance. Juniors often struggle to identify which resources are relevant and trustworthy. This creates inefficiency in preparation and reduces student confidence.

The Placement Bridge platform addresses these problems by creating a structured digital ecosystem where seniors can guide juniors through organized placement resources, interview questions, mentorship support, and knowledge-sharing modules. The system ensures that students receive authentic, reliable, and easily accessible guidance for effective placement preparation.

1.3 LEARNING CAPABILITIES AND DEVELOPMENTAL NEEDS

The Placement Bridge platform is designed not only as a placement guidance system but also as a collaborative learning environment that supports continuous knowledge enhancement and skill development. The platform improves the learning capabilities of students by enabling structured access to real-world placement experiences and educational resources.

One of the major learning capabilities of the platform is **experience-based learning**. Junior students can learn directly from seniors who have successfully participated in placement drives. This allows juniors to understand practical interview scenarios, preparation strategies, technical expectations, and common mistakes. Learning from real experiences helps students prepare more effectively compared to theoretical guidance alone.

Another important capability is **self-paced learning**. Students can access interview questions, guidance videos, quizzes, and programming-related content at their convenience. This flexibility allows learners to revise concepts repeatedly and improve their understanding without time limitations.

The system also supports **interactive learning** through discussion forums and query-resolution modules. Juniors can ask doubts related to coding, aptitude, interview preparation, resume building, or company-specific processes, while seniors provide guidance based on their practical experiences. This creates an engaging peer-learning environment.

The platform enhances **technical learning capabilities** by providing programming language-wise most asked questions and technical interview preparation materials. Students can strengthen their conceptual understanding in languages such as:

- C Programming
- C++
- Java
- Python
- JavaScript

The quiz and self-assessment modules further improve learning efficiency by helping students evaluate their preparation level. Instant feedback and result analysis encourage students to identify weak areas and improve continuously.

Additionally, the platform promotes:

- Collaborative learning
- Knowledge sharing
- Practical understanding of placement processes
- Confidence building
- Communication and mentorship culture

The Placement Bridge platform therefore acts as both a preparation system and an educational support platform that enhances the academic and professional learning capabilities of students.

1.4 GROWTH-ORIENTED STORYTELLING STRATEGIES

Growth orientation refers to the ability of a system to support long-term academic, professional, and institutional development. The Placement Bridge platform is designed with a strong growth-oriented approach that benefits students as well as educational institutions.

From the student perspective, the platform encourages continuous learning and skill improvement. Juniors gain exposure to real interview experiences, company-specific

preparation techniques, and technical guidance from seniors. This helps students develop confidence, improve problem-solving abilities, and become more placement-ready.

The mentorship structure of the platform creates a culture of collaborative growth. Senior students act as mentors who guide juniors through placement challenges, while juniors actively learn and contribute to discussions. This interaction promotes teamwork, communication, and peer support within the academic environment.

The project also contributes to career growth by helping students:

- Understand industry expectations
- Improve interview preparation strategies
- Develop technical and aptitude skills
- Access structured learning resources
- Build confidence for recruitment processes

From an institutional perspective, improved placement preparation can positively impact overall placement performance and institutional reputation. Better placement records strengthen the credibility of the institution and increase student satisfaction.

The platform is also growth-oriented from a technological perspective. The modular architecture of the system allows future expansion and integration of advanced functionalities such as:

- AI-based recommendation systems
- Personalized preparation pathways
- Analytics and performance tracking
- Resume analysis modules
- Real-time mentoring systems
- Multi-college integration

The system encourages equal learning opportunities for all students by providing centralized access to guidance and resources. This reduces dependency on personal networks and creates a more inclusive academic environment.

Overall, the Growth Orientation Study highlights that Placement Bridge is not only a placement preparation platform but also a long-term educational support system that promotes academic excellence, career readiness, and collaborative student development.

1.5 PROJECT SCOPE AND RELEVANCE

The scope of the Placement Bridge project is focused on developing a structured web-based platform that supports placement preparation through senior–junior interaction, mentorship, and knowledge sharing. The project aims to create an organized digital environment where students can access authentic placement-related guidance and resources.

The platform mainly focuses on the following areas:

- Placement guidance and mentorship
- Sharing of placement experiences
- Company-wise interview preparation
- Programming language interview questions
- Quiz and self-assessment activities
- Discussion and query resolution
- Structured access to preparation resources

The system allows senior students to upload guidance videos, share preparation strategies, and provide insights into recruitment processes. Junior students can explore this content according to their interests, targeted companies, or technical domains.

The project scope also includes the development of a secure and user-friendly platform using frontend, backend, and database technologies. The platform supports smooth interaction between users and ensures reliable data storage and retrieval.

The current version of the project is limited to academic placement preparation within a college environment. The platform is not designed for direct recruitment or job application processing. Instead, its primary purpose is to simplify placement preparation and improve student readiness.

Future scope of the project includes:

- AI-driven recommendation systems
- Skill assessment analytics
- Resume-building assistance
- Integration with training programs
- Institution-wide or multi-college deployment
- Mobile application support
- Personalized learning dashboards

The system can also be extended into a comprehensive career guidance platform that supports internships, higher education guidance, and industry mentorship programs.

Thus, the Placement Bridge project provides a scalable and future-ready foundation for improving placement preparation, mentorship, and collaborative learning within academic institutions.

CHAPTER 2

FEASIBILITY STUDY & PROJECT APPROACH

Feasibility study is an essential phase in the software development life cycle that helps determine whether a proposed project is practical, achievable, and beneficial within available resources and constraints. It evaluates the viability of the system from multiple perspectives such as technical capability, operational efficiency, economic affordability, and social impact. Conducting a feasibility study ensures that the project can be implemented successfully without major risks or limitations.

For the **Placement Bridge** – Senior–Junior Connection Platform, a detailed feasibility analysis has been carried out to examine whether the system can effectively fulfill its objectives of improving placement preparation, promoting mentorship, and enabling structured knowledge sharing among students. The study confirms that the project is realistic, cost-effective, technically achievable, and highly useful within the academic environment.

The feasibility analysis of the project is divided into the following categories:

- Technical Feasibility
- Economic Feasibility
- Operational Feasibility
- Social Feasibility

Each feasibility aspect helps in understanding the strengths, practicality, and future sustainability of the proposed system.

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2.1 Technical Feasibility

Technical feasibility evaluates whether the proposed system can be developed and implemented using the available technologies, tools, and technical expertise. It determines if the technical requirements of the project are achievable within the given resources and whether the development team possesses the required knowledge and skills.

The Placement Bridge platform is technically feasible because it is built using widely accepted and modern web development technologies that are reliable, scalable, and easy to maintain. The frontend of the system is developed using HTML, CSS, and JavaScript, which help in creating an interactive and responsive user interface. The backend is implemented using Node.js and Express.js, which provide efficient server-side processing and request handling. MongoDB/MySQL is used for managing structured and unstructured placement-related data.

These technologies are open-source, well-documented, and supported by large developer communities. This makes implementation easier and reduces technical complexity during development and maintenance. The development tools required for the project, such as Visual Studio Code, GitHub, and database management tools, are also easily available and commonly used in academic and professional environments.

The project does not require highly advanced hardware infrastructure or specialized software systems. Standard computing devices with internet connectivity are sufficient for development, testing, and deployment. Since the platform is web-based, users can access it from different devices through modern web browsers without installing additional software.

The modular architecture of the system further enhances technical feasibility. Different modules such as guidance videos, interview questions, quizzes, and discussions are designed independently while remaining interconnected. This structure simplifies debugging, testing, maintenance, and future upgrades.

The technical feasibility of the Placement Bridge platform is supported by the following factors:

- Availability of standard web technologies
- Ease of development and implementation
- Scalable and modular architecture
- Availability of technical documentation and resources
- Familiarity of developers with the selected technologies
- Minimal hardware requirements
- Compatibility across different operating systems and browsers

The system is therefore technically achievable, reliable, and suitable for long-term deployment within the academic environment.

2.2 Economic Feasibility

Economic feasibility analyzes the financial viability of the proposed system and determines whether the project can be developed and maintained within reasonable cost limits. It evaluates the balance between project expenses and expected benefits.

The Placement Bridge project is economically feasible because it utilizes open-source technologies and freely available development tools, significantly reducing the overall development cost. Technologies such as HTML, CSS, JavaScript, Node.js, Express.js, and MongoDB/MySQL do not require expensive licensing fees, making the system affordable for academic implementation.

The hardware requirements of the project are also minimal. Development and deployment can be performed using standard computers available in educational institutions. Since the application is web-based, there is no requirement for expensive infrastructure or dedicated high-end servers during the initial stages of deployment. Affordable cloud hosting or local hosting solutions can be used to run the system effectively.

The maintenance cost of the system is relatively low because of the modular design and use of commonly available technologies. Future updates and feature enhancements can be implemented without rebuilding the entire system.

The project provides several long-term benefits that justify the development effort and cost. By improving placement preparation and increasing student confidence, the system can contribute positively to placement performance and institutional reputation. Students can save time and effort by accessing authentic and structured guidance through a single platform instead of relying on scattered online resources.

Economic benefits of the system include:

- Reduced dependency on paid preparation platforms
- Lower software and infrastructure costs
- Cost-effective maintenance and scalability
- Improved student placement outcomes
- Better utilization of institutional academic resources
- Long-term sustainability through open-source technologies

The project therefore offers a high benefit-to-cost ratio and can be considered economically practical and sustainable for academic institutions.

2.3 Operational Feasibility

Operational feasibility evaluates how effectively the proposed system will function in a real-world environment and whether users will be able to operate it efficiently. It focuses on system usability, user acceptance, and the ability of the project to integrate smoothly within existing workflows.

The Placement Bridge platform is highly operationally feasible because it is specifically designed for students and follows a simple, user-friendly, and intuitive workflow. The interface is structured in a way that users can easily navigate through different modules such as placement guidance videos, interview questions, quizzes, and discussion forums without requiring technical training.

The platform integrates naturally within the academic ecosystem because placement preparation is already an important part of student life. The system enhances existing preparation practices by organizing resources and enabling structured mentorship rather than replacing current learning methods.

The operational workflow of the system is straightforward. Students can log in securely, access the dashboard, browse preparation resources, participate in discussions, and attempt quizzes. Senior students can contribute by sharing guidance videos and experiences. This interaction creates an efficient and collaborative learning environment.

The platform is designed to support smooth communication and knowledge sharing among students. Since most students are already familiar with web-based applications and online learning systems, adapting to the platform becomes easy and convenient.

Operational feasibility is strengthened through:

- User-friendly interface design
- Simple navigation and accessibility
- Minimal training requirements
- Structured organization of content
- Smooth integration into academic activities
- High usability and student engagement
- Efficient handling of placement-related resources

The system is expected to receive positive acceptance from both seniors and juniors because it directly addresses common placement preparation challenges and provides practical solutions.

Thus, the Placement Bridge platform is operationally feasible and suitable for implementation within educational institutions.

2.4 Social Feasibility

Social feasibility evaluates the impact of the proposed system on society, user behavior, collaboration, and overall community development. It examines whether the system is socially acceptable and whether it contributes positively to users and the environment in which it operates.

The Placement Bridge platform has strong social feasibility because it promotes collaboration, mentorship, equal learning opportunities, and peer-to-peer guidance among students. The platform creates a positive educational environment where knowledge and experiences are shared openly for the benefit of the student community.

One of the major social advantages of the system is that it reduces the communication gap between seniors and juniors. Senior students can guide juniors using their practical placement experiences, while juniors gain confidence and clarity regarding placement preparation. This mentorship culture encourages cooperation and mutual support within the institution.

The platform also helps reduce placement-related stress and anxiety by providing structured guidance and authentic information. Students often feel overwhelmed during placement season due to uncertainty and lack of preparation direction. Placement Bridge minimizes these issues by offering reliable and organized preparation resources.

The system promotes inclusivity by ensuring equal access to placement guidance for all students regardless of their social connections or academic background. Every student can benefit from the same resources and mentorship opportunities available on the platform.

Social benefits of the system include:

- Encouragement of peer learning and collaboration
- Improved student confidence and motivation
- Reduction in placement-related confusion and stress
- Equal access to placement preparation resources
- Promotion of mentorship culture within institutions
- Development of communication and teamwork skills
- Positive academic and career-oriented environment

The project also aligns with broader educational and social goals such as improving quality education, supporting career development, and encouraging knowledge sharing.

Overall, the Placement Bridge platform is socially beneficial, highly acceptable, and capable of creating a positive impact on the academic community by strengthening student collaboration and career readiness.

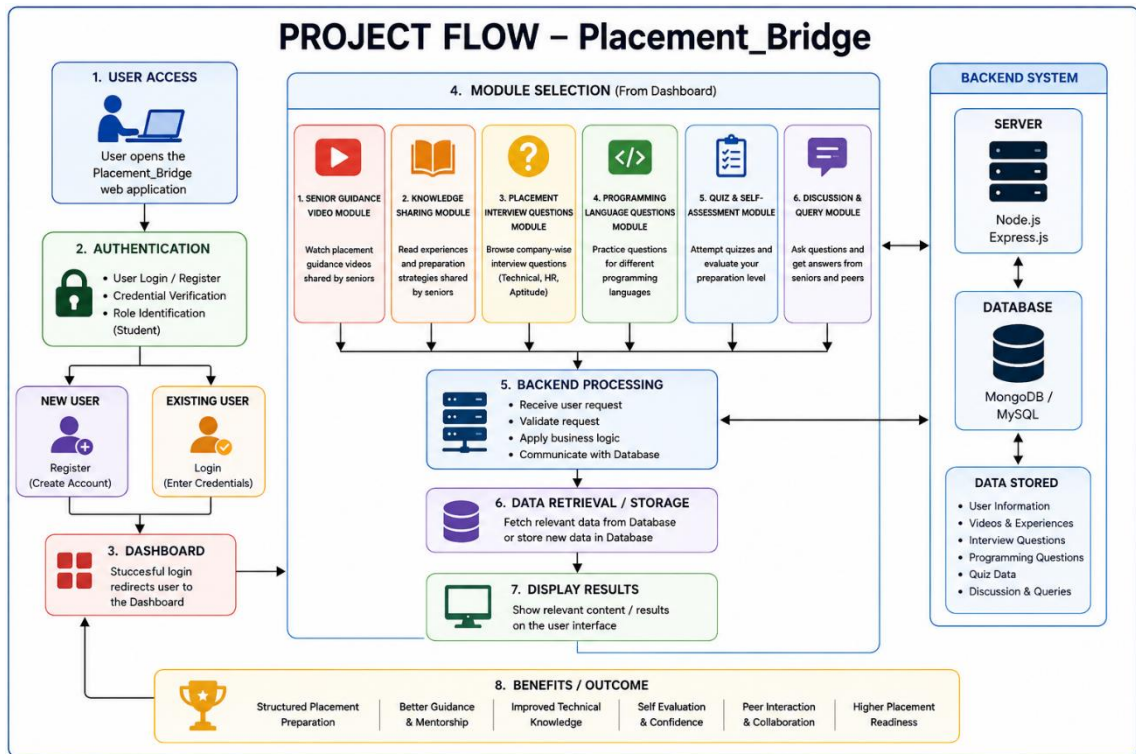


Figure 1: Project Flow

CHAPTER 3

Project Objective

3.1 PROJECT OBJECTIVE

Project objectives define the overall goals and intended outcomes of a software system. They provide a clear direction for development and help ensure that all modules and functionalities are aligned with the purpose of the project. The objectives of a project also help in identifying the problems to be solved and the expected benefits for users.

The **Placement Bridge** – Senior–Junior Connection Platform is developed with the primary objective of creating a structured, reliable, and collaborative placement preparation environment for students. The project aims to bridge the communication gap between senior and junior students by enabling organized knowledge sharing, mentorship, and access to authentic placement resources.

In today's competitive academic and professional environment, students often face difficulties during placement preparation due to scattered resources, lack of guidance, and insufficient awareness regarding interview processes. The Placement Bridge platform is designed to overcome these challenges by providing a centralized digital ecosystem where seniors can share their experiences and juniors can learn in a systematic and efficient manner.

The project focuses not only on technical implementation but also on improving student confidence, collaboration, and placement readiness. The objectives of the project are explained below in detail.

Main Objective of the Project

The primary objective of the Placement Bridge platform is to establish a strong and structured connection between senior and junior students for placement preparation and career guidance. The system aims to simplify the preparation process by organizing placement-related information and making it easily accessible to all students within the academic institution.

The project seeks to create a platform where seniors who have already participated in placement drives can share their practical experiences, preparation methods, interview

patterns, and technical guidance. Juniors can use this information to understand company expectations and prepare effectively for upcoming recruitment opportunities.

The system is designed to improve placement readiness, reduce confusion, and create a supportive mentorship environment that promotes collaborative learning and peer guidance.

Objectives of the Placement Bridge Platform

1. To Provide Structured Placement Guidance

One of the major objectives of the project is to organize placement-related resources in a structured and systematic manner. Many students face difficulty in identifying relevant preparation materials because online resources are scattered and unorganized.

The Placement Bridge platform aims to solve this issue by categorizing resources such as:

- Company-wise interview questions
- Technical preparation materials
- Guidance videos
- Placement experiences
- Programming language questions
- Quiz and assessment modules

This structured organization helps students focus on targeted preparation instead of wasting time searching for scattered information.

2. To Bridge the Gap Between Seniors and Juniors

The platform is developed to reduce the communication gap between experienced senior students and juniors who are beginning their placement journey. Seniors possess valuable practical knowledge regarding recruitment processes, coding rounds, aptitude tests, and HR interviews.

The objective of the project is to create a dedicated platform where this knowledge can be shared efficiently and systematically. Juniors can directly learn from seniors who have already faced placement interviews, making preparation more realistic and practical.

The project encourages:

- Peer mentorship
- Experience sharing
- Collaborative learning
- Guidance-based preparation

This interaction creates a stronger academic support system within the institution.

3. To Improve Placement Preparation Efficiency

Another important objective is to improve the efficiency of placement preparation by providing students with authentic and relevant guidance. The platform minimizes dependency on unreliable online resources and helps students prepare according to actual company recruitment patterns.

The project aims to save students' time and effort by providing all essential placement-related resources at a single platform. Students can access preparation content according to their targeted companies, technical domains, or learning requirements.

This objective helps students:

- Prepare in a focused manner
- Understand interview trends
- Improve technical knowledge
- Enhance confidence during interviews

4. To Promote Knowledge Sharing and Peer Learning

The Placement Bridge platform encourages students to actively participate in knowledge sharing and collaborative learning. Seniors contribute by uploading guidance videos, preparation strategies, and interview experiences, while juniors engage through discussions and query-resolution activities.

The objective is to create a continuous learning environment where students learn from one another rather than depending only on traditional classroom teaching. This promotes academic interaction and strengthens the culture of mentorship within the institution.

The platform supports:

- Discussion forums
- Query-resolution systems
- Shared preparation resources
- Collaborative problem-solving

This peer-learning model helps students improve both technical and communication skills.

5. To Provide Company-Specific Interview Preparation

Placement preparation varies according to company requirements, recruitment patterns, and job roles. Therefore, one of the key objectives of the project is to provide company-oriented preparation support.

The platform organizes interview questions and preparation experiences according to specific companies and technical domains. This helps students understand:

- Company interview patterns

- Frequently asked questions
- Technical expectations
- HR interview strategies
- Coding round preparation methods

Company-specific preparation increases the effectiveness of learning and improves placement performance.

6. To Support Technical Skill Development

The platform also focuses on strengthening technical concepts and programming knowledge required during placement interviews. The Programming Language Questions Module helps students practice and revise commonly asked technical questions in different languages.

The objective is to improve students' technical competency in areas such as:

- C Programming
- C++
- Java
- Python
- JavaScript
- Data Structures and Algorithms

The platform enables students to strengthen conceptual understanding and improve coding interview readiness.

7. To Enable Self-Assessment and Performance Evaluation

The Placement Bridge system includes quizzes and self-assessment modules that allow students to evaluate their preparation level.

The objective of this feature is to:

- Measure student understanding
- Identify strengths and weaknesses
- Encourage regular practice
- Improve confidence through assessment

Automated quiz evaluation helps students receive instant feedback and track their preparation progress effectively.

8. To Ensure Easy Accessibility and User-Friendly Interaction

Another important objective of the project is to provide a simple, responsive, and user-friendly interface so that students can use the system easily without technical difficulties.

The platform is designed with:

- Easy navigation
- Responsive layouts
- Clear module categorization
- Accessible dashboards
- Simple interaction workflows

This objective ensures that users can focus on learning and preparation rather than system complexity.

9. To Develop a Scalable and Future-Ready System

The Placement Bridge platform is developed using a modular and scalable architecture so that future enhancements can be integrated easily.

The project aims to support future extensions such as:

- AI-based recommendation systems
- Resume analysis tools
- Performance analytics
- Personalized preparation roadmaps
- Institution-wide deployment
- Mobile application integration

The system is therefore designed not only for present requirements but also for long-term growth and scalability.

CHAPTER 4

Hardware and Software Requirements

4.1 HARDWARE REQUIREMENTS

Hardware requirements refer to the physical components necessary for developing and running the Placement Bridge platform effectively. The project does not require specialized hardware infrastructure because it is built using lightweight web technologies and follows a client–server architecture. Standard computing systems with moderate configurations are sufficient for development and deployment.

The hardware configuration should support multitasking, database operations, frontend rendering, backend execution, and testing activities without affecting system performance. The following hardware components are recommended for smooth functioning of the system.

Processor

A system with an Intel Core i5 processor or higher is recommended for the development and execution of the Placement Bridge platform. The processor is responsible for handling backend operations, request processing, server execution, and multitasking activities during development.

A higher processing capability improves system responsiveness and ensures efficient handling of multiple modules simultaneously.

Recommended Configuration:

- Intel Core i5 / AMD Ryzen 5 or higher
- Multi-core processor support

RAM (Random Access Memory)

A minimum of **8 GB RAM** is required for smooth development and execution of the platform. RAM plays an important role in supporting simultaneous execution of development tools, backend servers, databases, browsers, and testing environments.

Adequate memory ensures better system responsiveness and minimizes performance delays during coding, debugging, and database interaction.

Recommended Configuration:

- Minimum: 8 GB RAM
- Preferred: 16 GB RAM for enhanced multitasking

Storage Requirements

The system requires sufficient storage space for storing source code, databases, project files, server configurations, uploaded resources, and development tools.

A **256 GB SSD (Solid State Drive)** or higher is recommended because SSD storage provides faster read/write operations, improving overall system performance and reducing loading times.

Recommended Configuration:

- Minimum: 256 GB SSD
- Preferred: 512 GB SSD or higher

Display and Resolution

A standard monitor with appropriate screen resolution is required for designing and testing the user interface of the Placement Bridge platform. Higher resolution improves visibility and supports responsive UI testing across different layouts.

Recommended Configuration:

- Minimum Resolution: 1366 × 768 pixels
- Preferred Resolution: Full HD (1920 × 1080)

Input Devices

Basic input devices such as a keyboard and mouse are required for coding, navigation, testing, and system interaction during development.

Required Devices:

- Standard keyboard
- Optical mouse or touchpad

Network Requirements

A stable internet connection is necessary for:

- Accessing online development resources
- Hosting and deployment
- Uploading guidance videos
- Version control using GitHub
- API testing and integration
- Database synchronization and updates

Internet connectivity also ensures smooth interaction between frontend and backend services during testing and deployment phases.

Recommended Configuration:

- Broadband/Wi-Fi connection
- Minimum speed: 10 Mbps

4.2 SOFTWARE REQUIREMENTS

Software requirements refer to the operating systems, programming languages, frameworks, databases, browsers, and development tools used to design, develop, execute, and maintain the Placement Bridge platform.

The software stack for the project is selected based on performance, scalability, ease of development, and compatibility with modern web application standards.

OperatingSystem

The Placement Bridge platform can be developed and executed on multiple operating systems because the selected technologies are cross-platform compatible.

Supported Operating Systems:

- Windows 10/11

- Linux (Ubuntu or similar distributions)
- macOS

These operating systems support Node.js, databases, development tools, and web browsers required for project implementation.

Frontend Technologies

The frontend is responsible for designing the visual interface and user interaction layer of the system. Modern web technologies are used to create a responsive, interactive, and user-friendly platform.

Technologies Used:

HTML (HyperText Markup Language)

HTML is used to structure web pages and organize content such as forms, dashboards, navigation menus, and module layouts.

CSS (Cascading Style Sheets)

CSS is used for styling the interface, improving layout design, responsiveness, colors, typography, and overall visual appearance.

JavaScript

JavaScript provides dynamic functionality and interactive behavior to the system such as form validation, dynamic content loading, video handling, and user interactions.

These technologies together create a responsive and engaging user interface for students.

Backend Technologies

The backend is responsible for handling business logic, server operations, authentication, request processing, and communication with the database.

Technologies Used:

Node.js

Node.js is used as the runtime environment for executing server-side JavaScript code efficiently. It supports asynchronous processing and improves application performance.

Express.js

Express.js is a lightweight Node.js framework used for creating RESTful APIs, routing, middleware integration, and backend request handling.

The backend architecture ensures efficient communication between users and system modules.

Database Management System

The database is responsible for storing and managing placement-related information in a structured manner.

Database Technologies Used:

- MongoDB / MySQL

The database stores:

- User information
- Placement experiences
- Guidance video metadata
- Interview questions
- Quiz data
- Discussion queries and responses

The database design supports efficient retrieval, updating, and management of data while ensuring consistency and reliability.

Development Tools

Several development tools are used during coding, debugging, testing, and collaboration.

Tools Used:

Visual Studio Code (VS Code)

Used as the primary code editor for writing and managing frontend and backend code.

GitHub

Used for version control, code collaboration, project management, and maintaining development history.

MySQL Workbench / MongoDB Compass

Used for database management, query execution, and database monitoring.

Postman

Used for API testing and backend request validation.

These tools improve development efficiency and simplify debugging and maintenance processes.

Server Environment

The Placement Bridge platform requires a server environment for hosting backend services and APIs.

Server Technologies:

- Node Server
- Apache Server (optional during deployment)
- Localhost environment during development

The server handles user requests, data communication, and module execution efficiently.

Web Browsers

Modern web browsers are required for testing and accessing the platform.

Supported Browsers:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge

These browsers ensure compatibility, responsiveness, and smooth user interaction.

CHAPTER 5

Project Flow

Project flow describes the overall working structure of the system and explains how different components interact with each other to perform specific tasks. It provides a clear understanding of how data moves through the system, how users interact with different modules, and how the backend processes request to generate desired outputs.

For the **Placement Bridge** – Senior–Junior Connection Platform, the project flow is designed to ensure smooth navigation, structured placement preparation, and efficient knowledge sharing between users. The system follows a modular workflow approach, where each module performs a specific function while remaining connected to the overall architecture.

The flow begins with user access and continues through various modules such as guidance videos, interview questions, programming preparation, quizzes, and discussion forums. All modules interact with the backend server and database to store and retrieve data efficiently.

Overview of Project Flow

The Placement Bridge platform operates on a **client–server architecture**, where:

- The **frontend (client side)** handles user interaction and displays content
- The **backend (server side)** processes requests and applies business logic
- The **database** stores and retrieves placement-related data

The workflow ensures that users can easily access relevant information, interact with the system, and benefit from structured learning resources.

The overall system flow can be summarized as follows:

1. User accesses the platform
2. Authentication and verification
3. Dashboard navigation
4. Module selection
5. Backend request processing
6. Data retrieval from database
7. Display of relevant content

8. User interaction and feedback

5.1 System Entry and Access Flow

The project flow begins when a user (student) accesses the Placement Bridge platform through a web browser. The homepage or landing page is loaded, which acts as the entry point of the system.

The system then ensures secure access through authentication mechanisms. Users must log in using valid credentials to access protected features such as guidance videos, quizzes, and discussion forums. This ensures data security and prevents unauthorized usage.

Once authentication is successful, the user is redirected to the main dashboard.

The entry flow includes:

- Opening the web application
- Loading the homepage interface
- User login or registration
- Credential verification
- Redirection to dashboard

This initial stage ensures that only authorized users can interact with the platform and access placement-related resources.

5.2 Dashboard and Navigation Flow

The dashboard serves as the central control panel of the Placement Bridge platform. It provides access to all modules and acts as the main interface for user interaction.

The dashboard is designed with simplicity and clarity so that students can easily navigate through different sections without confusion. It displays various modules such as guidance videos, placement questions, programming questions, quizzes, and knowledge-sharing sections.

When a user selects a module, a request is sent to the backend server, which processes the request and retrieves relevant data from the database. The requested module is then displayed on the user interface.

This flow ensures smooth navigation and efficient access to different system features.

5.3 Senior Guidance Video Module Flow

The Senior Guidance Video Module is one of the core components of the platform. It allows senior students to share their placement experiences through recorded videos, providing juniors with real-world insights into the recruitment process.

In this module, senior students upload videos along with relevant details such as company name, job role, and technology stack. The backend processes this information and stores the video metadata in the database.

Junior students can browse or search videos based on their interests or targeted companies. Once a video is selected, it is streamed through the platform for viewing.

This module creates a virtual mentorship environment where juniors learn directly from seniors.

5.4 Knowledge Sharing Module Flow

The Knowledge Sharing Module enables seniors to share written experiences, preparation strategies, interview processes, and useful insights. This module focuses on structured content rather than video-based guidance.

When a senior submits content, the backend validates the data and stores it in the database. The content is categorized based on company, topic, or preparation type.

Junior students can browse, search, and read this information to enhance their preparation. The structured format ensures that knowledge is easily accessible and reusable.

This module helps preserve valuable placement knowledge and promotes continuous learning within the student community.

5.5 Placement Interview Questions Module Flow

This module provides company-wise and category-wise interview questions that are commonly asked during placement processes. It helps students understand interview patterns and prepare accordingly.

The system stores questions in different categories such as:

- Technical questions
- HR interview questions
- Aptitude questions

When a user selects a company or category, the backend retrieves relevant questions from the database and displays them in an organized format.

Students can use this module to practice and improve their understanding of interview expectations.

5.6 Programming Language Questions Module Flow

The Programming Language Module focuses on strengthening core technical skills required for coding interviews. It provides frequently asked questions for different programming languages and topics.

Students can select a programming language such as C, C++, Java, Python, or JavaScript. The backend then fetches relevant questions from the database and displays them in a structured manner.

This module supports revision, concept clarity, and technical preparation. It helps students improve their coding knowledge and problem-solving skills.

5.7 Quiz and Self-Assessment Module Flow

The Quiz Module allows students to evaluate their preparation level through interactive assessments. It provides topic-wise or language-wise quizzes that test students' understanding of key concepts.

When a student selects a quiz, the system retrieves questions from the database and displays them on the interface. After the student submits answers, the backend evaluates the responses and generates results instantly.

The results help students identify strengths and weaknesses, encouraging continuous improvement.

This module supports self-learning and performance evaluation.

5.8 Discussion and Query Resolution Flow

The Discussion Module enables interaction between students for doubt clarification and knowledge sharing. Junior students can post queries related to placement preparation, technical concepts, or interview strategies.

These queries are stored in the database, and senior students or other users can respond with appropriate guidance. All responses are visible to users, making the discussion beneficial for multiple students.

This module promotes peer learning, collaboration, and active engagement among users.

5.9 Backend and Database Interaction Flow

The backend plays a crucial role in managing all system operations. It acts as a bridge between the frontend and the database. Every user action, such as selecting a module, submitting data, or requesting information, is processed by the backend.

The backend performs the following functions:

- Validates user requests
- Applies business logic
- Communicates with the database
- Retrieves or stores data
- Sends responses to the frontend

The database stores structured data related to:

- User profiles
- Placement experiences
- Guidance videos
- Interview questions
- Quiz data
- Discussion queries

Efficient backend and database interaction ensures system reliability, data consistency, and smooth performance.

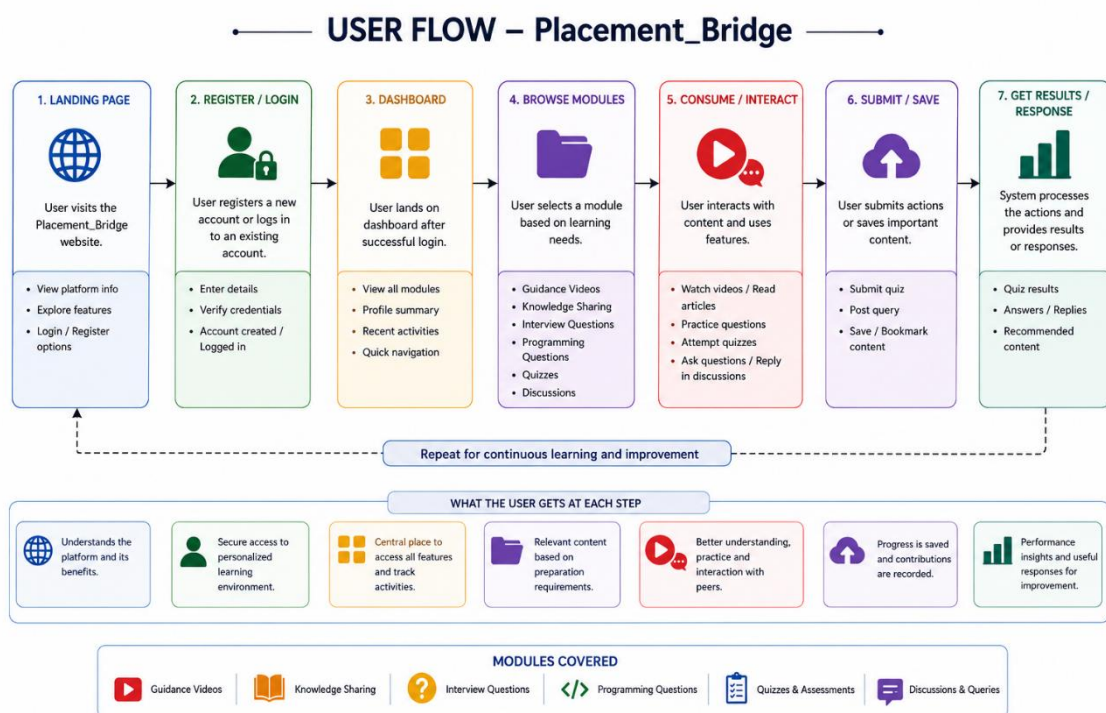


Figure 2: User Flow Diagram

CHAPTER 6

Project Outcome

Project outcome represents the final results, achievements, and impact of the developed system after its successful implementation. It highlights how effectively the project meets its objectives and solves the identified problems. The outcome also reflects the benefits gained by users and the overall contribution of the system to the intended environment.

The **Placement Bridge platform** has been successfully developed as a structured, user-friendly, and efficient web-based system that supports placement preparation through mentorship, knowledge sharing, and interactive learning. The system achieves its goal of bridging the gap between experienced senior students and juniors by providing a centralized platform for guidance and collaboration.

Overview of Project Outcome

The outcome of the Placement Bridge project demonstrates that the platform effectively simplifies placement preparation by organizing resources, enabling peer learning, and providing real-world insights into recruitment processes.

The system ensures that students no longer need to depend on scattered online resources or informal communication channels. Instead, they can access all essential placement-related information in one place, which improves preparation efficiency and reduces confusion.

The platform successfully integrates multiple modules such as guidance videos, interview questions, programming preparation, quizzes, and discussions into a single cohesive system. Each module contributes to enhancing the overall learning experience and placement readiness of students.

Key Outcomes of the Project

1. Centralized Placement Preparation Platform

One of the major outcomes of the project is the development of a centralized system where students can access all placement-related resources in an organized manner.

The platform provides:

- Company-specific interview questions
- Senior guidance videos
- Knowledge-sharing content
- Programming practice questions
- Quiz and self-assessment modules
- Discussion and query-resolution system

This centralization eliminates the need to search across multiple platforms and improves preparation efficiency.

2. Improved Student Guidance and Mentorship

The platform successfully establishes a structured mentorship system where senior students guide juniors through their placement journey.

Seniors share:

- Real interview experiences
- Preparation strategies
- Common mistakes to avoid
- Company-specific insights

This direct interaction helps juniors gain practical knowledge and reduces uncertainty during placement preparation.

3. Enhanced Learning Experience

The Placement Bridge platform improves the learning experience by combining theoretical knowledge with practical insights.

Students benefit from:

- Experience-based learning
- Self-paced study
- Interactive discussions
- Continuous feedback through quizzes

This integrated learning approach helps students understand concepts more effectively and apply them in real-world scenarios.

4. Strengthened Technical and Problem-Solving Skills

Through the Programming Questions Module and Interview Questions Module, students are able to enhance their technical knowledge and problem-solving abilities.

The system supports preparation in areas such as:

- Programming languages (C, C++, Java, Python, JavaScript)
- Data Structures and Algorithms
- Technical interview questions
- Aptitude and logical reasoning

Regular practice and structured content help students improve their coding skills and interview performance.

5. Self-Assessment and Performance Improvement

The quiz and assessment module enables students to evaluate their preparation level and track their progress.

Students can:

- Attempt quizzes based on different topics
- Receive instant results and feedback
- Identify strengths and weak areas
- Improve performance through repeated practice

This outcome encourages continuous learning and helps students build confidence over time.

6. Increased Student Confidence and Placement Readiness

One of the most important outcomes of the project is the improvement in student confidence and readiness for placement processes.

By accessing real interview experiences and structured guidance, students gain clarity about:

- Recruitment processes
- Interview patterns
- Technical expectations
- Preparation strategies

This reduces anxiety and helps students approach placement drives with better confidence and preparation.

7. Promotion of Collaborative Learning Environment

The platform successfully promotes a culture of collaboration and peer learning within the academic environment.

Students actively participate in:

- Knowledge sharing
- Discussion forums
- Query resolution
- Experience exchange

This collaborative approach enhances communication skills, teamwork, and mutual support among students.

8. Efficient and Scalable System Implementation

The project demonstrates successful implementation of a scalable and modular web-based system.

Key technical achievements include:

- Smooth frontend and backend integration
- Efficient database management
- Secure user authentication
- Modular system architecture
- Easy navigation and user-friendly interface

The system is designed in a way that allows future enhancements without major modifications.

Future Enhancements

The Placement Bridge platform has strong potential for future development and expansion. Some possible enhancements include:

- AI-based recommendation systems for personalized learning
- Resume-building and analysis tools
- Performance analytics and progress tracking dashboards
- Integration with online coding platforms
- Mobile application development
- Multi-institution or nationwide deployment

- Real-time mentorship and chat features

These improvements can further enhance the effectiveness and usability of the platform.

6.10. Implementation of Code:

```
import { motion } from "framer-motion";

import { Button } from "@/components/ui-elements";

import { useLocation } from "wouter";

import { ArrowRight, Briefcase, GraduationCap, ShieldCheck, Zap, Video,
MessageSquare, UserCircle, Users, FileText } from "lucide-react";

export default function Landing() {

  const [, setLocation] = useLocation();

  return (

    <div className="flex flex-col items-center">

      {/* Hero Section */}

      <section className="w-full min-h-[90vh] flex items-center justify-center relative
overflow-hidden px-4">

        <div className="absolute inset-0 z-0">

          {/* landing page hero futuristic neon tech workspace */}

          <div className="absolute inset-0 bg-gradient-to-b from-background/40 via-
background/80 to-background" />

        </div>
```

```
<div className="max-w-5xl mx-auto relative z-10 text-center flex flex-col items-center">
```

```
  <motion.div
```

```
    initial={{ opacity: 0, y: 20 }}
```

```
    animate={{ opacity: 1, y: 0 }}
```

```
    transition={{ duration: 0.6 }}
```

```
    className="inline-flex items-center rounded-full border border-primary/30 bg-primary/10 px-4 py-1.5 text-sm font-medium text-primary mb-8 backdrop-blur-sm shadow-[0_0_15px_rgba(0,255,255,0.2)]"
```

```
  >
```

```
    <Zap className="mr-2 h-4 w-4" /> The Future of Recruitment
```

```
</motion.div>
```

```
<motion.h1
```

```
  initial={{ opacity: 0, y: 20 }}
```

```
  animate={{ opacity: 1, y: 0 }}
```

```
  transition={{ duration: 0.6, delay: 0.1 }}
```

```
  className="text-5xl md:text-7xl lg:text-8xl font-black tracking-tighter mb-6"
```

```
>
```

```
  Bridge the gap to your <br />
```

```
  <span className="text-gradient">Dream Career</span>
```

```
</motion.h1>
```

```
<motion.p
```

```
  initial={{ opacity: 0, y: 20 }}
```

```
  animate={{ opacity: 1, y: 0 }}
```

```
  transition={{ duration: 0.6, delay: 0.2 }}
```

```

        className="text-lg md:text-2xl text-muted-foreground max-w-2xl mb-10"
      >
        A high-performance platform connecting visionary students with industry-leading
        employers. Fast, secure, and beautifully designed.
      </motion.p>

      <motion.div
        initial={{ opacity: 0, y: 20 }}
        animate={{ opacity: 1, y: 0 }}
        transition={{ duration: 0.6, delay: 0.3 }}
        className="flex flex-col sm:flex-row gap-4 w-full justify-center"
      >
        <Button size="lg" onClick={() => setLocation("/auth")} className="text-lg px-
8">
          Join the Network <ArrowRight className="ml-2 w-5 h-5" />
        </Button>

        <Button variant="outline" size="lg" onClick={() => setLocation("/auth")}
className="text-lg px-8">
          Post a Job
        </Button>
      </motion.div>
    </div>
  </section>

  { /* Features Section */ }

  <section className="w-full py-24 px-4 bg-black/40 relative z-10">
    <div className="max-w-7xl mx-auto">
      <div className="text-center mb-16">

```

```
<h2 className="text-3xl md:text-5xl font-bold mb-4">Why Choose <span
className="text-primary">PlacementBridge</span>?</h2>
```

```
<p className="text-muted-foreground max-w-2xl mx-auto">Engineered for
speed, built for success. We've redefined how universities and companies handle
recruitment.</p>
```

```
</div>
```

```
<div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-8">
```

```
{[
```

```
  { icon: ShieldCheck, title: "Authentication & Secure Login", desc: "Safe and
secure access for all members of the community." },
```

```
  { icon: Video, title: "Senior Video Uploads", desc: "Watch placement
experiences and tech stack insights from alumni." },
```

```
  { icon: MessageSquare, title: "Interview Questions", desc: "Access company-
wise interview questions shared by seniors." },
```

```
  { icon: UserCircle, title: "Personal Profiles", desc: "Showcase your placement
details or career interests." },
```

```
  { icon: Users, title: "Q&A Discussion Forum", desc: "Direct interaction between
juniors and seniors for query resolution." },
```

```
  { icon: FileText, title: "Resource Sharing", desc: "A centralized hub for notes,
links, and study channels." }
```

```
].map((feature, i) => (
```

```
<motion.div
```

```
  key={i}
```

```
  initial={{ opacity: 0, y: 20 }}
```

```
  whileInView={{ opacity: 1, y: 0 }}
```

```
  viewport={{ once: true }}
```

```
  transition={{ delay: i * 0.1 }}
```

```
  className="glass-panel p-8 rounded-3xl neon-border group cursor-default"
```

```
>
```

```

        <div className="w-14 h-14 rounded-2xl bg-primary/20 flex items-center
justify-center mb-6 group-hover:bg-primary transition-colors duration-500">

            <feature.icon className="w-7 h-7 text-primary group-hover:text-
background transition-colors duration-500" />

        </div>

        <h3 className="text-2xl font-bold mb-3">{feature.title}</h3>

        <p className="text-muted-foreground leading-relaxed">{feature.desc}</p>

    </motion.div>

    )}

</div>

</div>

</section>

</div>

);

}

import { useApplications, useUpdateApplicationStatus } from "@/hooks/use-
applications";

import { useJobs } from "@/hooks/use-jobs";

import { useAuth } from "@/hooks/use-auth";

import { Card, Button, Badge } from "@/components/ui-elements";

import { format } from "date-fns";

import { Check, X, FileText, Mail, User } from "lucide-react";

export default function EmployerApplicationsPage() {

    const { user } = useAuth();

    const { data: applications, isLoading: appsLoading } = useApplications();

    const { data: jobs, isLoading: jobsLoading } = useJobs();

    const updateStatus = useUpdateApplicationStatus();

```

```
if (appsLoading || jobsLoading) return <div className="p-8 text-center animate-pulse">Loading candidates...</div>;
```

```
// Filter jobs owned by employer
```

```
const myJobIds = new Set(jobs?.filter(j => j.employerId === user?.id).map(j => j.id) || []);
```

```
// Filter applications for those jobs
```

```
const myApplications = applications?.filter(app => myJobIds.has(app.jobId)) || [];
```

```
return (
```

```
<div className="max-w-7xl mx-auto px-4 py-8 w-full">
```

```
<div className="mb-10">
```

```
<h1 className="text-4xl font-bold mb-2">Candidate Applications</h1>
```

```
<p className="text-muted-foreground">Review and manage candidates applying for your roles.</p>
```

```
</div>
```

```
<div className="space-y-6">
```

```
{myApplications.length > 0 ? myApplications.map(app => {
```

```
  const job = jobs?.find(j => j.id === app.jobId);
```

```
  return (
```

```
<Card key={app.id} className="p-6 md:p-8 border-l-4 border-l-primary flex flex-col md:flex-row gap-6">
```

```
<div className="flex-1">
```

```
<div className="flex items-center gap-3 mb-4">
```

```

        <Badge variant={app.status === 'accepted' ? 'success' : app.status ===
'rejected' ? 'danger' : 'warning'} className="uppercase">

            {app.status}

        </Badge>

        <span className="text-sm text-muted-foreground">

            Applied on {app.appliedAt ? format(new Date(app.appliedAt), 'MMM d,
yyyy') : 'Unknown'}

        </span>

    </div>

    <h3 className="text-2xl font-bold mb-1 flex items-center">

        <User className="w-5 h-5 mr-2 text-primary" />

        {app.student?.fullName || "Candidate"}

    </h3>

    <p className="text-muted-foreground text-sm mb-4">

        Applying for <strong className="text-foreground">{job?.title || "Unknown
Role"}</strong>

    </p>

    <div className="bg-black/30 rounded-xl p-4 mb-4 border border-white/5">

        <h4 className="text-sm font-semibold text-primary mb-2 flex items-
center">

            <FileText className="w-4 h-4 mr-2" /> Candidate Profile

        </h4>

        <p      className="text-sm      text-muted-foreground      mb-2"><span
className="text-white">Bio:</span> {app.student?.bio || "No bio provided."}</p>

        <p className="text-sm text-muted-foreground"><span className="text-
white">Skills:</span> {app.student?.skills || "Not specified."}</p>

    </div>

```

```
</div>
```

```
<div className="flex flex-row md:flex-col gap-3 justify-center md:min-w-[160px]">
```

```
<Button
```

```
  variant="outline"
```

```
  className="w-full justify-start"
```

```
  onClick={() => alert('Feature coming soon: Contact candidate')}
```

```
>
```

```
<Mail className="w-4 h-4 mr-2" /> Contact
```

```
</Button>
```

```
{app.status === 'pending' && (
```

```
<>
```

```
<Button
```

```
  variant="primary"
```

```
  className="w-full justify-start bg-green-500 hover:bg-green-600 shadow-[0_0_15px_rgba(34,197,94,0.3)] text-white"
```

```
  isLoading={updateStatus.isPending}
```

```
  onClick={() => updateStatus.mutate({ id: app.id, status: 'accepted' })}
```

```
>
```

```
<Check className="w-4 h-4 mr-2" /> Accept
```

```
</Button>
```

```
<Button
```

```
  variant="danger"
```

```
  className="w-full justify-start"
```

```
  isLoading={updateStatus.isPending}
```

```
  onClick={() => updateStatus.mutate({ id: app.id, status: 'rejected' })}
```

```
>
```

```

        <X className="w-4 h-4 mr-2" /> Reject
      </Button>
    </>
  )}
</div>
</Card>

);

}): (
  <div className="text-center py-20 glass-panel rounded-2xl border border-dashed
border-white/10">
    <Users className="w-12 h-12 text-muted-foreground mx-auto mb-4 opacity-50"
/>

    <h3 className="text-xl font-medium mb-2">No applications yet</h3>

    <p className="text-muted-foreground">When candidates apply to your jobs,
they will appear here.</p>
  </div>
)}
</div>
</div>

);
}

import { useJobs, useCreateJob } from "@/hooks/use-jobs";
import { useAuth } from "@/hooks/use-auth";
import { Card, Button, Input, Textarea, Label } from "@/components/ui-elements";
import { Plus, Users, Briefcase } from "lucide-react";
import { useState } from "react";
import { motion, AnimatePresence } from "framer-motion";

```

```

export default function EmployerDashboard() {

  const { user } = useAuth();

  const { data: jobs, isLoading: jobsLoading } = useJobs();

  const createJobMutation = useCreateJob();

  const [isCreating, setIsCreating] = useState(false);


  const [formData, setFormData] = useState({

    title: "",

    company: user?.fullName || "",

    location: "",

    salary: "",

    description: "",

    requirements: ""

  });


  const myJobs = jobs?.filter(j => j.employerId === user?.id) || [];


  const handleCreateJob = async (e: React.FormEvent) => {

    e.preventDefault();

    await createJobMutation.mutateAsync(formData);

    setIsCreating(false);

    setFormData({ title: "", company: user?.fullName || "", location: "", salary: "",
description: "", requirements: "" });

  };


  if (jobsLoading) return <div className="p-8 text-center">Loading...</div>;

```

```

return (

  <div className="max-w-7xl mx-auto px-4 py-8 w-full relative">

    <div className="flex flex-col md:flex-row justify-between items-start md:items-center mb-10 gap-4">

      <div>

        <h1 className="text-4xl font-bold mb-2">Employer Dashboard</h1>

        <p className="text-muted-foreground">Manage your job postings and find the perfect candidates.</p>

      </div>

      <Button onClick={() => setIsCreating(true)} size="lg">

        <Plus className="w-5 h-5 mr-2" /> Post New Job

      </Button>

    </div>

    <AnimatePresence>

      {isCreating && (

        <motion.div

          initial={{ opacity: 0, y: -20 }}

          animate={{ opacity: 1, y: 0 }}

          exit={{ opacity: 0, height: 0 }}

          className="mb-10"

        >

          <Card className="p-8 border-primary/50 shadow-[0_0_30px_rgba(0,255,255,0.1)]">

            <h2 className="text-2xl font-bold mb-6 text-primary">Create Job Posting</h2>

            <form onSubmit={handleCreateJob} className="space-y-6">

              <div className="grid grid-cols-1 md:grid-cols-2 gap-6">

```

```

<div className="space-y-2">

  <Label>Job Title</Label>

  <Input    required    value={formData.title}    onChange={e    =>
setFormData({...formData, title: e.target.value})} placeholder="Software Engineer" />

</div>

<div className="space-y-2">

  <Label>Company Name</Label>

  <Input    required    value={formData.company}    onChange={e    =>
setFormData({...formData, company: e.target.value})} />

</div>

<div className="space-y-2">

  <Label>Location</Label>

  <Input    required    value={formData.location}    onChange={e    =>
setFormData({...formData, location: e.target.value})} placeholder="Remote / New York,
NY" />

</div>

<div className="space-y-2">

  <Label>Salary Range (Optional)</Label>

  <Input          value={formData.salary}          onChange={e          =>
setFormData({...formData, salary: e.target.value})} placeholder="$100k - $130k" />

</div>

</div>

<div className="space-y-2">

  <Label>Job Description</Label>

  <Textarea    required    value={formData.description}    onChange={e    =>
setFormData({...formData, description: e.target.value})} className="min-h-[150px]" />

</div>

<div className="space-y-2">

  <Label>Requirements</Label>

```

```

        <Textarea required value={formData.requirements} onChange={e =>
setFormData({...formData, requirements: e.target.value})} placeholder="List skills and
qualifications..." />

```

```

    </div>

```

```

    <div className="flex justify-end space-x-4">

```

```

        <Button type="button" variant="ghost" onClick={() =>
setIsCreating(false)}>Cancel</Button>

```

```

        <Button type="submit" isLoading={createJobMutation.isPending}>Publish
Job</Button>

```

```

    </div>

```

```

  </form>

```

```

</Card>

```

```

</motion.div>

```

```

  )}

```

```

</AnimatePresence>

```

```

<h3 className="text-2xl font-bold mb-6">Your Active Postings</h3>

```

```

<div className="grid grid-cols-1 md:grid-cols-2 gap-6">

```

```

  {myJobs.length > 0 ? myJobs.map(job => (

```

```

    <Card key={job.id} className="p-6">

```

```

      <div className="flex justify-between items-start mb-4">

```

```

        <div>

```

```

          <h4 className="text-xl font-bold text-white">{job.title}</h4>

```

```

          <p className="text-primary text-sm">{job.location}</p>

```

```

        </div>

```

```

      <div className="w-10 h-10 rounded-full bg-primary/10 flex items-center
justify-center text-primary">

```

```

        <Briefcase className="w-5 h-5" />

```

```

      </div>

```

```

    </div>

    <p className="text-sm text-muted-foreground line-clamp-2 mb-6">{job.description}</p>

    <div className="flex justify-between items-center pt-4 border-t border-white/5">

      <span className="text-sm text-muted-foreground flex items-center">

        <Users className="w-4 h-4 mr-2" /> Applications

      </span>

      <Button variant="outline" size="sm" onClick={() => window.location.href = '/employer/applications'}>

        View Candidates

      </Button>

    </div>

  </Card>

  )) : (

    <div className="col-span-full py-16 text-center border border-dashed border-white/10 rounded-2xl glass-panel">

      <Briefcase className="w-12 h-12 text-muted-foreground mx-auto mb-4 opacity-50" />

      <h3 className="text-xl font-medium mb-2">No active postings</h3>

      <p className="text-muted-foreground mb-4">You haven't posted any jobs yet.</p>

      <Button onClick={() => setIsCreating(true)}>Create Your First Job</Button>

    </div>

  )}

</div>

</div>

);

}

```

```

import { motion } from "framer-motion";

import { Button } from "@components/ui-elements";

import { useLocation } from "wouter";

import { ArrowRight, Briefcase, GraduationCap, ShieldCheck, Zap, Video,
MessageSquare, UserCircle, Users, FileText } from "lucide-react";

export default function Landing() {

  const [, setLocation] = useLocation();

  return (

    <div className="flex flex-col items-center">

      {/* Hero Section */}

      <section className="w-full min-h-[90vh] flex items-center justify-center relative
overflow-hidden px-4">

        <div className="absolute inset-0 z-0">

          {/* landing page hero futuristic neon tech workspace */}

          <div className="absolute inset-0 bg-gradient-to-b from-background/40 via-
background/80 to-background" />

        </div>

        <div className="max-w-5xl mx-auto relative z-10 text-center flex flex-col items-
center">

          <motion.div

```

```
initial={{ opacity: 0, y: 20 }}
```

```
animate={{ opacity: 1, y: 0 }}
```

```
transition={{ duration: 0.6 }}
```

```
className="inline-flex items-center rounded-full border border-primary/30 bg-  
primary/10 px-4 py-1.5 text-sm font-medium text-primary mb-8 backdrop-blur-sm  
shadow-[0_0_15px_rgba(0,255,255,0.2)]"
```

```
>
```

```
<Zap className="mr-2 h-4 w-4" /> The Future of Recruitment
```

```
</motion.div>
```

```
<motion.h1
```

```
initial={{ opacity: 0, y: 20 }}
```

```
animate={{ opacity: 1, y: 0 }}
```

```
transition={{ duration: 0.6, delay: 0.1 }}
```

```
className="text-5xl md:text-7xl lg:text-8xl font-black tracking-tighter mb-6"
```

```
>
```

```
Bridge the gap to your <br />
```

```
<span className="text-gradient">Dream Career</span>
```

```
</motion.h1>
```

```
<motion.p
```

```
initial={{ opacity: 0, y: 20 }}
```

```
animate={{ opacity: 1, y: 0 }}
```

```
transition={{ duration: 0.6, delay: 0.2 }}
```

```
className="text-lg md:text-2xl text-muted-foreground max-w-2xl mb-10"
```

```
>
```

A high-performance platform connecting visionary students with industry-leading employers. Fast, secure, and beautifully designed.

```

</motion.p>

<motion.div
  initial={{ opacity: 0, y: 20 }}
  animate={{ opacity: 1, y: 0 }}
  transition={{ duration: 0.6, delay: 0.3 }}
  className="flex flex-col sm:flex-row gap-4 w-full justify-center"
>
  <Button size="lg" onClick={() => setLocation("/auth")} className="text-lg px-
8">
    Join the Network <ArrowRight className="ml-2 w-5 h-5" />
  </Button>

  <Button variant="outline" size="lg" onClick={() => setLocation("/auth")}
className="text-lg px-8">
    Post a Job
  </Button>
</motion.div>
</div>
</section>

{/* Features Section */}
<section className="w-full py-24 px-4 bg-black/40 relative z-10">
  <div className="max-w-7xl mx-auto">
    <div className="text-center mb-16">
      <h2 className="text-3xl md:text-5xl font-bold mb-4">Why Choose <span
className="text-primary">PlacementBridge</span>?</h2>
      <p className="text-muted-foreground max-w-2xl mx-auto">Engineered for
speed, built for success. We've redefined how universities and companies handle
recruitment.</p>

```

</div>

```
<div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-8">

  {[

    { icon: ShieldCheck, title: "Authentication & Secure Login", desc: "Safe and
secure access for all members of the community." },

    { icon: Video, title: "Senior Video Uploads", desc: "Watch placement
experiences and tech stack insights from alumni." },

    { icon: MessageSquare, title: "Interview Questions", desc: "Access company-
wise interview questions shared by seniors." },

    { icon: UserCircle, title: "Personal Profiles", desc: "Showcase your placement
details or career interests." },

    { icon: Users, title: "Q&A Discussion Forum", desc: "Direct interaction between
juniors and seniors for query resolution." },

    { icon: FileText, title: "Resource Sharing", desc: "A centralized hub for notes,
links, and study channels." }

  ]}.map((feature, i) => (

    <motion.div

      key={i}

      initial={{ opacity: 0, y: 20 }}

      whileInView={{ opacity: 1, y: 0 }}

      viewport={{ once: true }}

      transition={{ delay: i * 0.1 }}

      className="glass-panel p-8 rounded-3xl neon-border group cursor-default"

    >

      <div className="w-14 h-14 rounded-2xl bg-primary/20 flex items-center
justify-center mb-6 group-hover:bg-primary transition-colors duration-500">

        <feature.icon className="w-7 h-7 text-primary group-hover:text-
background transition-colors duration-500" />

      </div>
```

```

        <h3 className="text-2xl font-bold mb-3">{feature.title}</h3>

        <p className="text-muted-foreground leading-relaxed">{feature.desc}</p>

    </motion.div>

    )})
</div>

</div>

</section>

</div>

);
}

import { useResources } from "@/hooks/use-resources";
import { useAuth } from "@/hooks/use-auth";
import { useState } from "react";
import { useLocation } from "wouter";
import { Button, Input, Textarea } from "@/components/ui-elements";
import { Card, CardContent, CardHeader, CardTitle } from "@/components/ui/card";
import { Badge } from "@/components/ui/badge";
import { FileText, Link as LinkIcon, Video, Plus, ExternalLink } from "lucide-react";
import { formatDistanceToNow } from "date-fns";
import { Dialog, DialogContent, DialogHeader, DialogTitle, DialogTrigger } from
"@/components/ui/dialog";
import { Select, SelectContent, SelectItem, SelectTrigger, SelectValue } from
"@/components/ui/select";

export default function ResourcesPage() {
    const { resources = [], create, isLoading } = useResources();
    const { user } = useAuth();
    const [, setLocation] = useLocation();

```

```
const [isOpen, setIsOpen] = useState(false);
```

```
const [formData, setFormData] = useState({
```

```
  title: "",
```

```
  content: "",
```

```
  type: "note",
```

```
  company: ""
```

```
});
```

```
const handleCreate = async () => {
```

```
  await create(formData);
```

```
  setIsOpen(false);
```

```
  setFormData({ title: "", content: "", type: "note", company: "" });
```

```
};
```

```
const getIcon = (type: string) => {
```

```
  switch (type) {
```

```
    case 'video': return <Video className="w-5 h-5 text-red-400" />;
```

```
    case 'link': return <LinkIcon className="w-5 h-5 text-blue-400" />;
```

```
    default: return <FileText className="w-5 h-5 text-yellow-400" />;
```

```
  }
```

```
};
```

```
if (isLoading) return <div className="p-8 text-center text-muted-foreground">Loading  
resources...</div>;
```

```
return (
```

```
  <div className="max-w-6xl mx-auto px-4 py-12 space-y-8 w-full">
```

```

<div className="flex items-center justify-between">

  <div className="space-y-1">

    <h1 className="text-3xl font-display font-bold text-white">Resource
    Sharing</h1>

    <p className="text-muted-foreground">Access notes, interview questions, and
    senior placement videos.</p>

  </div>

```

```

<Dialog open={isOpen} onOpenChange={(open) => {
  if (open && !user) {
    setLocation("/auth");
    return;
  }
  setIsOpen(open);
}}>

```

```

<DialogTrigger asChild>

```

```

  <Button>

```

```

    <Plus className="w-4 h-4 mr-2" /> Share Resource

```

```

  </Button>

```

```

</DialogTrigger>

```

```

<DialogContent className="glass-panel border-white/10 text-white">

```

```

  <DialogHeader>

```

```

    <DialogTitle>Share a New Resource</DialogTitle>

```

```

  </DialogHeader>

```

```

  <div className="space-y-4 pt-4">

```

```

    <div className="space-y-2">

```

```

      <label className="text-sm font-medium">Title</label>

```

```

      <Input

```

```

      value={formData.title}

      onChange={e => setFormData(prev => ({ ...prev, title: e.target.value })))}

      className="bg-white/5 border-white/10"

    />
  </div>

  <div className="space-y-2">

    <label className="text-sm font-medium">Type</label>

    <Select value={formData.type} onValueChange={v => setFormData(prev =>
    ({ ...prev, type: v })))>

      <SelectTrigger className="bg-white/5 border-white/10">

        <SelectValue />

      </SelectTrigger>

      <SelectContent className="glass-panel border-white/10 text-white">

        <SelectItem value="note">Note / Interview Q&A</SelectItem>

        <SelectItem value="link">Useful Link</SelectItem>

        <SelectItem value="video">Placement Video</SelectItem>

      </SelectContent>

    </Select>

  </div>

  <div className="space-y-2">

    <label className="text-sm font-medium">Link or Content</label>

    <Textarea

      value={formData.content}

      onChange={e => setFormData(prev => ({ ...prev, content: e.target.value })))}

      className="bg-white/5 border-white/10"

    />

  </div>

```

```

<div className="space-y-2">
  <label className="text-sm font-medium">Company (Optional)</label>
  <Input
    value={formData.company}
    onChange={e => setFormData(prev => ({ ...prev, company: e.target.value
    ))}
    className="bg-white/5 border-white/10"
  />
</div>

<Button onClick={handleCreate} className="w-full">Share</Button>
</div>
</DialogContent>
</Dialog>
</div>

<div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6">
  {resources.map((r: any) => (
    <Card key={r.id} className="glass-panel border-white/5 hover-elevate
transition-all group">
      <CardHeader className="flex flex-row items-center space-x-3 space-y-0">
        <div className="w-10 h-10 rounded-lg bg-white/5 flex items-center justify-
center group-hover:bg-primary/20 transition-colors">
          {getIcon(r.type)}
        </div>
        <div className="flex-1 min-w-0">
          <CardTitle className="text-base font-bold text-white
truncate">{r.title}</CardTitle>
          <p className="text-xs text-muted-foreground truncate">

```

```

        by {r.author.fullName} • {formatDistanceToNow(new Date(r.createdAt))}
ago

    </p>

</div>

</CardHeader>

<CardContent className="space-y-4">

    {r.company && (

        <Badge variant="outline" className="border-primary/30 text-primary bg-
primary/5">

            {r.company}

        </Badge>

    )}

    <p className="text-sm text-muted-foreground line-clamp-3">

        {r.content}

    </p>

    {r.type === 'link' || r.type === 'video' ? (

        <a

            href={r.content.startsWith('http') ? r.content : `https://${r.content}`}

            target="_blank"

            rel="noopener noreferrer"

            className="flex items-center text-xs font-bold text-primary hover:underline"

        >

            View Resource <ExternalLink className="w-3 h-3 ml-1" />

        </a>

    ) : null}

</CardContent>

</Card>

)}}

```

```

    </div>
  </div>
);
}
import { useApplications } from "@/hooks/use-applications";
import { useJobs } from "@/hooks/use-jobs";
import { useAuth } from "@/hooks/use-auth";
import { Card, Badge, Button } from "@/components/ui-elements";
import { format } from "date-fns";
import { Link } from "wouter";
import { Briefcase, Building2, MapPin, Calendar, ExternalLink } from "lucide-react";

export default function StudentApplicationsPage() {
  const { user } = useAuth();

  const { data: applications, isLoading: appsLoading } = useApplications();
  const { data: jobs, isLoading: jobsLoading } = useJobs();

  if (appsLoading || jobsLoading) return <div className="p-8 text-center animate-pulse">Loading your applications...</div>;

  // Filter to only show the student's own applications
  const myApplications = applications?.filter(app => app.studentId === user?.id) || [];

  return (
    <div className="max-w-7xl mx-auto px-4 py-8 w-full">
      <div className="mb-10">
        <h1 className="text-4xl font-bold mb-2">My Applications</h1>

```

```
<p className="text-muted-foreground">Track the status of roles you've applied for.</p>
```

```
</div>
```

```
<div className="grid grid-cols-1 gap-6">
```

```
{myApplications.length > 0 ? myApplications.map(app => {
```

```
  const job = jobs?.find(j => j.id === app.jobId);
```

```
  return (
```

```
    <Card key={app.id} className="p-6 flex flex-col md:flex-row items-start md:items-center justify-between gap-6 transition-all hover:bg-white/[0.02]">
```

```
      <div className="flex items-start gap-6">
```

```
        <div className="hidden md:flex w-16 h-16 rounded-2xl bg-gradient-to-br from-primary/20 to-accent/20 border border-white/10 items-center justify-center">
```

```
          <Briefcase className="w-8 h-8 text-white/80" />
```

```
        </div>
```

```
      <div>
```

```
        <h3 className="text-2xl font-bold mb-2">{job?.title || "Unknown Role"}</h3>
```

```
        <div className="flex flex-wrap items-center gap-4 text-sm text-muted-foreground mb-4">
```

```
          <span className="flex items-center text-primary font-medium">
```

```
            <Building2 className="w-4 h-4 mr-1.5" /> {job?.company || "Unknown Company"}
```

```
          </span>
```

```
          <span className="flex items-center">
```

```
            <MapPin className="w-4 h-4 mr-1.5" /> {job?.location || "N/A"}
```

```
          </span>
```

```
          <span className="flex items-center">
```

```

        <Calendar className="w-4 h-4 mr-1.5" /> Applied {app.appliedAt ?
format(new Date(app.appliedAt), 'MMM d, yyyy') : 'Unknown'}

    </span>

</div>

</div>

</div>

<div className="flex flex-row md:flex-col items-center md:items-end justify-
between w-full md:w-auto gap-4">

    <Badge

        variant={app.status === 'accepted' ? 'success' : app.status === 'rejected' ?
'danger' : 'warning'}

        className="px-4 py-1.5 text-sm uppercase tracking-wider"

    >

        {app.status}

    </Badge>

    <Link href="/dashboard" className="text-sm text-primary hover:underline
flex items-center">

        Find more jobs <ExternalLink className="w-3 h-3 ml-1" />

    </Link>

</div>

</Card>

);

}): (

    <div className="text-center py-20 glass-panel rounded-2xl border border-dashed
border-white/10">

        <Briefcase className="w-12 h-12 text-muted-foreground mx-auto mb-4 opacity-
50" />

        <h3 className="text-xl font-medium mb-2">No applications found</h3>

```

```
    <p className="text-muted-foreground mb-6">You haven't applied to any jobs yet. Start exploring!</p>
```

```
    <Link href="/dashboard">
```

```
      <Button>Browse Jobs</Button>
```

```
    </Link>
```

```
  </div>
```

```
  )}
```

```
</div>
```

```
</div>
```

```
);
```

```
}
```

```
import { useJobs } from "@/hooks/use-jobs";
```

```
import { useApply, useApplications } from "@/hooks/use-applications";
```

```
import { Card, Button, Badge } from "@/components/ui-elements";
```

```
import { Building2, MapPin, DollarSign, Clock, Search } from "lucide-react";
```

```
import { format } from "date-fns";
```

```
import { useState } from "react";
```

```
import { Input } from "@/components/ui-elements";
```

```
export default function StudentDashboard() {
```

```
  const { data: jobs, isLoading: jobsLoading } = useJobs();
```

```
  const { data: applications } = useApplications();
```

```
  const applyMutation = useApply();
```

```
  const [searchTerm, setSearchTerm] = useState("");
```

```
  if (jobsLoading) return <div className="p-8 text-center"><span className="animate-pulse">Loading jobs...</span></div>;
```

```

const appliedJobIds = new Set(applications?.map(app => app.jobId) || []);

const filteredJobs = jobs?.filter(job =>
  job.title.toLowerCase().includes(searchTerm.toLowerCase()) ||
  job.company.toLowerCase().includes(searchTerm.toLowerCase())
);

return (
  <div className="max-w-7xl mx-auto px-4 py-8 w-full">
    <div className="flex flex-col md:flex-row justify-between items-center mb-10 gap-4">
      <div>
        <h1 className="text-4xl font-bold mb-2">Discover Opportunities</h1>
        <p className="text-muted-foreground">Find and apply for the best roles matching your skills.</p>
      </div>
      <div className="relative w-full md:w-96">
        <Search className="absolute left-3 top-1/2 -translate-y-1/2 w-5 h-5 text-muted-foreground" />
        <Input
          placeholder="Search roles, companies..."
          className="pl-10"
          value={searchTerm}
          onChange={(e) => setSearchTerm(e.target.value)}
        />
      </div>
    </div>
  </div>
)

```

```

<div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6">
  {filteredJobs?.map(job => (
    <Card key={job.id} className="flex flex-col h-full">
      <div className="p-6 flex-1">
        <div className="flex justify-between items-start mb-4">
          <div className="w-12 h-12 rounded-xl bg-white/5 border border-white/10
flex items-center justify-center">
            <Building2 className="w-6 h-6 text-primary" />
          </div>
          {appliedJobIds.has(job.id) ? (
            <Badge variant="success">Applied</Badge>
          ) : (
            <Badge variant="outline">New</Badge>
          )}
        </div>
        <h3 className="text-xl font-bold mb-1">{job.title}</h3>
        <p className="text-primary font-medium mb-4">{job.company}</p>

        <div className="space-y-2 mb-6">
          <div className="flex items-center text-sm text-muted-foreground">
            <MapPin className="w-4 h-4 mr-2" /> {job.location}
          </div>
          {job.salary && (
            <div className="flex items-center text-sm text-muted-foreground">
              <DollarSign className="w-4 h-4 mr-2" /> {job.salary}
            </div>
          )}
        </div>
      </div>
    </Card>
  )}
</div>

```

```

        <div className="flex items-center text-sm text-muted-foreground">
            <Clock className="w-4 h-4 mr-2" /> {job.postedAt ? format(new
Date(job.postedAt), 'MMM d, yyyy') : 'Recently'}
        </div>
    </div>

    <div className="text-sm text-muted-foreground line-clamp-3 mb-4">
        {job.description}
    </div>
</div>

<div className="p-6 pt-0 mt-auto border-t border-white/5 flex gap-3 pt-4">
    <Button
        className="w-full"
        variant={appliedJobIds.has(job.id) ? "secondary" : "primary"}
        disabled={appliedJobIds.has(job.id) || applyMutation.isPending}
        onClick={() => applyMutation.mutate(job.id)}
    >
        {appliedJobIds.has(job.id) ? "Application Sent" : "Apply Now"}
    </Button>
</div>
</Card>
)}}

{filteredJobs?.length === 0 && (
    <div className="col-span-full text-center py-20 text-muted-foreground">
        No jobs found matching your criteria.
    </div>

```

```

    })
  </div>
</div>

);
}

@import
url('https://fonts.googleapis.com/css2?family=Outfit:wght@300;400;500;600;700&family=Plus+Jakarta+Sans:wght@400;500;600;700&display=swap');

@tailwind base;
@tailwind components;
@tailwind utilities;

@layer base {
  :root {
    --font-display: 'Outfit', sans-serif;
    --font-body: 'Plus Jakarta Sans', sans-serif;

    /* Dark futuristic palette */
    --background: 240 10% 4%;
    --foreground: 0 0% 98%;

    --card: 240 10% 6%;
    --card-foreground: 0 0% 98%;

    --popover: 240 10% 6%;
    --popover-foreground: 0 0% 98%;
  }
}

```

```

/* Cyan / Teal Neon */

--primary: 186 100% 50%;
--primary-foreground: 240 10% 4%;

--secondary: 240 10% 12%;
--secondary-foreground: 0 0% 98%;

--muted: 240 10% 12%;
--muted-foreground: 240 5% 65%;

/* Magenta / Purple Neon */

--accent: 280 100% 60%;
--accent-foreground: 0 0% 98%;

--destructive: 0 84% 60%;
--destructive-foreground: 0 0% 98%;

--border: 240 10% 16%;
--input: 240 10% 12%;
--ring: 186 100% 50%;

--radius: 0.75rem;
}

```

```

body {

  @apply bg-background text-foreground antialiased selection:bg-primary/30
  selection:text-primary;
}

```

```

font-family: var(--font-body);

/* Subtle grid background */

background-image:
  linear-gradient(to right, rgba(255,255,255,0.03) 1px, transparent 1px),
  linear-gradient(to bottom, rgba(255,255,255,0.03) 1px, transparent 1px);
background-size: 50px 50px;
background-position: center center;
background-attachment: fixed;
}

h1, h2, h3, h4, h5, h6 {
  font-family: var(--font-display);
  @apply tracking-tight;
}

}

@layer utilities {
  .glass-panel {
    @apply bg-card/60 backdrop-blur-xl border border-white/5 shadow-2xl;
  }

  .neon-border {
    @apply relative z-0;
  }

  .neon-border::before {
    content: "";

```

```

    @apply absolute -inset-[1px] rounded-[inherit] -z-10 bg-gradient-to-r from-primary
via-accent to-primary opacity-0 transition-opacity duration-500;
}

```

```

.neon-border:hover::before {
  @apply opacity-50 blur-sm;
}

```

```

.text-gradient {
  @apply bg-clip-text text-transparent bg-gradient-to-r from-primary to-accent;
}
}

```

```

import { Link, useLocation } from "wouter";
import { useAuth } from "@/hooks/use-auth";
import { Button } from "../ui-elements";
import { BriefcaseIcon, LayoutDashboard, LogOut, Menu, X } from "lucide-react";
import { useState } from "react";
import { motion, AnimatePresence } from "framer-motion";

```

```

export function Layout({ children }: { children: React.ReactNode }) {
  const { user, logout } = useAuth();
  const [, setLocation] = useLocation();
  const [isMobileMenuOpen, setIsMobileMenuOpen] = useState(false);

  const handleLogout = async () => {
    await logout();
    setLocation("/");
  }
}

```

```
};
```

```
const navLinks = user?.role === "employer"
```

```
  ? [
```

```
    { label: "My Postings", href: "/dashboard" },
```

```
    { label: "Applications", href: "/employer/applications" },
```

```
  ]
```

```
  : user?.role === "student"
```

```
  ? [
```

```
    { label: "Browse Jobs", href: "/dashboard" },
```

```
    { label: "My Applications", href: "/student/applications" },
```

```
    { label: "Q&A", href: "/qa" },
```

```
    { label: "Resources", href: "/resources" },
```

```
  ]
```

```
  : [
```

```
    { label: "Q&A", href: "/qa" },
```

```
    { label: "Resources", href: "/resources" },
```

```
  ];
```

```
return (
```

```
  <div className="min-h-screen flex flex-col bg-[#0a0f1a]">
```

```
    <header className="sticky top-0 z-50 w-full glass-panel border-b border-white/5">
```

```
      <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 h-20 flex items-center justify-between">
```

```
        <Link href="/" className="flex items-center space-x-3 group">
```

```
          <div className="w-10 h-10 rounded-xl bg-gradient-to-br from-primary to-accent flex items-center justify-center shadow-[0_0_15px_rgba(0,255,255,0.4)] group-hover:shadow-[0_0_25px_rgba(0,255,255,0.6)] transition-all">
```

```

        <BriefcaseIcon className="text-white w-6 h-6" />

    </div>

    <span className="font-display font-bold text-xl tracking-tight text-white group-
hover:text-transparent group-hover:bg-clip-text group-hover:bg-gradient-to-r group-
hover:from-primary group-hover:to-accent transition-all">

        PlacementBridge

    </span>

</Link>

{/* Desktop Nav */}

<nav className="hidden md:flex items-center space-x-8">

    {navLinks.map((link) => (

        <Link key={link.href} href={link.href} className="text-sm font-medium text-
muted-foreground hover:text-primary transition-colors">

            {link.label}

        </Link>

    ))}

    {user ? (

        <div className="flex items-center space-x-4 pl-8 border-l border-white/10">

            <div className="flex flex-col items-end">

                <span
                    className="text-sm
                    font-bold
                    text-
foreground">{user.fullName}</span>

                <span className="text-xs text-primary capitalize">{user.role}</span>

            </div>

            <Button
                variant="ghost"
                size="icon"
                onClick={handleLogout}
title="Logout">

                <Logout className="w-5 h-5" />

            </Button>

        </div>

    ) : (

```

```

    <>

    <Link href="/auth" className="text-sm font-medium text-muted-foreground
hover:text-primary transition-colors">Sign In</Link>

    <Button onClick={() => setLocation("/auth")}>Get Started</Button>

  </>

)}

</nav>

{/* Mobile Menu Toggle */}

<div className="md:hidden">

  <Button variant="ghost" size="icon" onClick={() =>
setIsMobileMenuOpen(!isMobileMenuOpen)}>

    {isMobileMenuOpen ? <X /> : <Menu />}

  </Button>

</div>

</div>

</header>

{/* Mobile Nav */}

<AnimatePresence>

  {isMobileMenuOpen && (

    <motion.div

      initial={{ opacity: 0, height: 0 }}

      animate={{ opacity: 1, height: "auto" }}

      exit={{ opacity: 0, height: 0 }}

      className="md:hidden glass-panel border-b border-white/5"

    >

      <div className="px-4 pt-2 pb-6 space-y-4 flex flex-col">

        {navLinks.map((link) => (

```

```

        <Link      key={link.href}      href={link.href}      onClick={()      =>
setIsMobileMenuOpen(false)}      className="block      py-2      text-muted-foreground
hover:text-primary">

          {link.label}

        </Link>

      )}

    {user ? (

      <>

        <div className="py-2 border-b border-white/10">

          <p className="font-bold text-foreground">{user.fullName}</p>

          <p className="text-sm text-primary capitalize">{user.role}</p>

        </div>

        <Button      variant="outline"      className="w-full      mt-4"
onClick={handleLogout}>

          <LogOut className="w-4 h-4 mr-2" /> Logout

        </Button>

      </>

    ) : (

      <>

        <Button variant="ghost" className="w-full justify-start" onClick={() => {
setLocation("/auth"); setIsMobileMenuOpen(false); }}>

          Sign In

        </Button>

        <Button className="w-full" onClick={() => { setLocation("/auth");
setIsMobileMenuOpen(false); }}>

          Get Started

        </Button>

      </>

    )}

```

```

    </div>

    </motion.div>

  )}

</AnimatePresence>

<main className="flex-1 flex flex-col relative">

  <div className="absolute top-0 left-1/4 w-96 h-96 bg-primary/20 rounded-full
blur-[128px] -z-10 pointer-events-none" />

  <div className="absolute bottom-0 right-1/4 w-96 h-96 bg-accent/20 rounded-full
blur-[128px] -z-10 pointer-events-none" />

  {children}

</main>

<footer className="glass-panel border-t border-white/5 py-12">

  <div className="max-w-7xl mx-auto px-4 grid grid-cols-1 md:grid-cols-4 gap-
12">

    <div className="space-y-4">

      <div className="flex items-center space-x-3">

        <div className="w-8 h-8 rounded-lg bg-gradient-to-br from-primary to-accent
flex items-center justify-center shadow-[0_0_10px_rgba(0,255,255,0.4)]">

          <BriefcaseIcon className="text-white w-5 h-5" />

        </div>

        <span      className="font-display      font-bold      text-lg      text-
white">PlacementBridge</span>

      </div>

      <p className="text-sm text-muted-foreground">

        Empowering students and alumni through mentorship, resources, and career
opportunities.

      </p>

    </div>

    <div className="space-y-4">

```

```

    <h4 className="font-bold text-white">Company</h4>

    <ul className="space-y-2 text-sm text-muted-foreground">

      <li><Link href="/about" className="hover:text-primary transition-
colors">About Us</Link></li>

      <li><Link href="/careers" className="hover:text-primary transition-
colors">Careers</Link></li>

      <li><Link href="/contact" className="hover:text-primary transition-
colors">Contact Us</Link></li>

      {user?.role === 'admin' && <li><Link href="/admin" className="hover:text-
primary transition-colors">Admin Panel</Link></li>}

    </ul>

  </div>

  <div className="space-y-4">

    <h4 className="font-bold text-white">Resources</h4>

    <ul className="space-y-2 text-sm text-muted-foreground">

      <li><Link href="/qa" className="hover:text-primary transition-colors">Q&A
Forum</Link></li>

      <li><Link href="/resources" className="hover:text-primary transition-
colors">Materials</Link></li>

      <li><Link href="/dashboard" className="hover:text-primary transition-
colors">Job Board</Link></li>

    </ul>

  </div>

  <div className="space-y-4">

    <h4 className="font-bold text-white">Social</h4>

    <div className="flex space-x-4">

      <a href="#" className="w-10 h-10 rounded-lg bg-white/5 flex items-center
justify-center hover:bg-primary/20 transition-all text-muted-foreground hover:text-
primary">

        <svg className="w-5 h-5" fill="currentColor" viewBox="0 0 24 24"><path
d="M19 0h-14c-2.761 0-5 2.239-5 5v14c0 2.761 2.239 5 5 5h14c2.762 0 5-2.239 5-5v-

```

14c0-2.761-2.238-5-5-5zm-11 19h-3v-11h3v11zm-1.5-12.268c-.966 0-1.75-.79-1.75-1.764s.784-1.764 1.75-1.764 1.75.79 1.75 1.764-.783 1.764-1.75 1.764zm13.5 12.268h-3v-5.604c0-3.368-4-3.113-4 0v5.604h-3v-11h3v1.765c1.396-2.586 7-2.777 7 2.476v6.759z"/></svg>

<svg className="w-5 h-5" fill="currentColor" viewBox="0 0 24 24"><path d="M24 4.557c-.883.392-1.832.656-2.828.775 1.017-.609 1.798-1.574 2.165-2.724-.951.564-2.005.974-3.127 1.195-.897-.957-2.178-1.555-3.594-1.555-3.179 0-5.515 2.966-4.797 6.045-4.091-.205-7.719-2.165-10.148-5.144-1.29 2.213-.669 5.108 1.523 6.574-.806-.026-1.566-.247-2.229-.616-.054 2.281 1.581 4.415 3.949 4.89-.693.188-1.452.232-2.224.084.626 1.956 2.444 3.379 4.6 3.419-2.07 1.623-4.678 2.348-7.29 2.04 2.179 1.397 4.768 2.212 7.548 2.212 9.142 0 14.307-7.721 13.995-14.646.962-.695 1.797-1.562 2.457-2.549z"/></svg>

<svg className="w-5 h-5" fill="currentColor" viewBox="0 0 24 24"><path d="M22.675 0h-21.35c-.732 0-1.325.593-1.325 1.325v21.351c0 .731.593 1.324 1.325 1.324h11.495v-9.294h-3.128v-3.622h3.128v-2.671c0-3.1 1.893-4.788 4.659-4.788 1.325 0 2.463.099 2.795.143v3.24l-1.918.001c-1.504 0-1.795.715-1.795 1.763v2.313h3.587l-.467 3.622h-3.12v9.293h6.116c.73 0 1.323-.593 1.323-1.325v-21.35c0-.732-.593-1.325-1.325-1.325z"/></svg>

</div>

</div>

</div>

<div className="max-w-7xl mx-auto px-4 mt-12 pt-8 border-t border-white/5 text-center text-xs text-muted-foreground">

© {new Date().getFullYear()} PlacementBridge. Connecting talent with opportunity.

</div>

```

    </footer>

  </div>

);
}

import React, { forwardRef } from "react";
import { Loader2 } from "lucide-react";
import { clsx, type ClassValue } from "clsx";
import { twMerge } from "tailwind-merge";

export function cn(...inputs: ClassValue[]) {
  return twMerge(clsx(inputs));
}

interface ButtonProps extends React.ButtonHTMLAttributes<HTMLButtonElement> {
  variant?: "primary" | "secondary" | "outline" | "ghost" | "danger";
  size?: "sm" | "md" | "lg" | "icon";
  isLoading?: boolean;
}

export const Button = forwardRef<HTMLButtonElement, ButtonProps>(
  ({ className, variant = "primary", size = "md", isLoading, children, disabled, ...props }, ref) => {
    const baseStyles = "inline-flex items-center justify-center rounded-xl font-medium transition-all duration-300 focus-visible:outline-none focus-visible:ring-2 focus-visible:ring-primary disabled:opacity-50 disabled:pointer-events-none active:scale-95";

    const variants = {

```

```

    primary: "bg-primary text-primary-foreground shadow-[0_0_15px_rgba(0,255,255,0.3)] hover:shadow-[0_0_25px_rgba(0,255,255,0.5)] hover:-translate-y-0.5",

    secondary: "bg-secondary text-secondary-foreground hover:bg-secondary/80",

    outline: "border border-primary/50 text-primary hover:bg-primary/10",

    ghost: "hover:bg-white/5 text-muted-foreground hover:text-foreground",

    danger: "bg-destructive text-destructive-foreground hover:bg-destructive/90 shadow-[0_0_15px_rgba(255,0,0,0.2)]",

  };

  const sizes = {

    sm: "h-9 px-4 text-xs",

    md: "h-11 px-6 text-sm",

    lg: "h-14 px-8 text-base",

    icon: "h-10 w-10",

  };

  return (

    <button

      ref={ref}

      className={cn(baseStyles, variants[variant], sizes[size], className)}

      disabled={disabled || isLoading}

      {...props}

    >

      {isLoading && <Loader2 className="mr-2 h-4 w-4 animate-spin" />}

      {children}

    </button>

  );

```

```

    }

    );

    Button.displayName = "Button";

export      const      Card      =      forwardRef<HTMLDivElement,
React.HTMLAttributes<HTMLDivElement>>(({ className, ...props }, ref) => (

    <div ref={ref} className={cn("glass-panel rounded-2xl neon-border overflow-
hidden", className)} {...props} />

));

Card.displayName = "Card";

export      const      Input      =      forwardRef<HTMLInputElement,
React.InputHTMLAttributes<HTMLInputElement>>(({ className, ...props }, ref) => (

    <input

        ref={ref}

        className={cn(

            "flex h-12 w-full rounded-xl border border-white/10 bg-black/40 px-4 py-2 text-sm
text-foreground ring-offset-background file:border-0 file:bg-transparent file:text-sm
file:font-medium placeholder:text-muted-foreground focus-visible:outline-none focus-
visible:ring-2 focus-visible:ring-primary focus-visible:border-transparent transition-all",

            className

        )}

        {...props}

    />

));

Input.displayName = "Input";

export      const      Textarea      =      forwardRef<HTMLTextAreaElement,
React.TextareaHTMLAttributes<HTMLTextAreaElement>>(({ className, ...props },
ref) => (

```

```

<textarea
  ref={ref}

  className={cn(
    "flex min-h-[100px] w-full rounded-xl border border-white/10 bg-black/40 px-4 py-3
    text-sm text-foreground ring-offset-background placeholder:text-muted-foreground
    focus-visible:outline-none focus-visible:ring-2 focus-visible:ring-primary focus-
    visible:border-transparent transition-all resize-y",
    className
  )}
  {...props}
/>
));

Textarea.displayName = "Textarea";

```

```

export const Label = forwardRef<HTMLLabelElement,
React.LabelHTMLAttributes<HTMLLabelElement>>(({ className, ...props }, ref) => (
  <label ref={ref} className={cn("text-sm font-medium leading-none peer-
  disabled:cursor-not-allowed peer-disabled:opacity-70", className)} {...props} />
));

Label.displayName = "Label";

```

```

export const Badge = ({ className, variant = "default", ...props }:
React.HTMLAttributes<HTMLDivElement> & { variant?: "default" | "success" |
"warning" | "danger" | "outline" }) => {
  const variants = {
    default: "bg-primary/20 text-primary border border-primary/30",
    success: "bg-green-500/20 text-green-400 border border-green-500/30",
    warning: "bg-yellow-500/20 text-yellow-400 border border-yellow-500/30",
    danger: "bg-red-500/20 text-red-400 border border-red-500/30",
    outline: "border border-white/20 text-foreground",
  }

```

```

};

return (
  <div className={cn("inline-flex items-center rounded-full px-2.5 py-0.5 text-xs font-
semibold transition-colors", variants[variant], className)} {...props} />
);
};

import type { Config } from "tailwindcss";

export default {
  darkMode: ["class"],
  content: ["/client/index.html", "/client/src/**/*.{js,jsx,ts,tsx}"],
  theme: {
    extend: {
      borderRadius: {
        lg: ".5625rem", /* 9px */
        md: ".375rem", /* 6px */
        sm: ".1875rem", /* 3px */
      },
      colors: {
        // Flat / base colors (regular buttons)
        background: "hsl(var(--background) / <alpha-value>)",
        foreground: "hsl(var(--foreground) / <alpha-value>)",
        border: "hsl(var(--border) / <alpha-value>)",
        input: "hsl(var(--input) / <alpha-value>)",
        card: {
          DEFAULT: "hsl(var(--card) / <alpha-value>)",
          foreground: "hsl(var(--card-foreground) / <alpha-value>)",

```

```

border: "hsl(var(--card-border) / <alpha-value>)",
},
popover: {
  DEFAULT: "hsl(var(--popover) / <alpha-value>)",
  foreground: "hsl(var(--popover-foreground) / <alpha-value>)",
  border: "hsl(var(--popover-border) / <alpha-value>)",
},
primary: {
  DEFAULT: "hsl(var(--primary) / <alpha-value>)",
  foreground: "hsl(var(--primary-foreground) / <alpha-value>)",
  border: "var(--primary-border)",
},
secondary: {
  DEFAULT: "hsl(var(--secondary) / <alpha-value>)",
  foreground: "hsl(var(--secondary-foreground) / <alpha-value>)",
  border: "var(--secondary-border)",
},
muted: {
  DEFAULT: "hsl(var(--muted) / <alpha-value>)",
  foreground: "hsl(var(--muted-foreground) / <alpha-value>)",
  border: "var(--muted-border)",
},
accent: {
  DEFAULT: "hsl(var(--accent) / <alpha-value>)",
  foreground: "hsl(var(--accent-foreground) / <alpha-value>)",
  border: "var(--accent-border)",
},

```

```

destructive: {
  DEFAULT: "hsl(var(--destructive) / <alpha-value>)",
  foreground: "hsl(var(--destructive-foreground) / <alpha-value>)",
  border: "var(--destructive-border)",
},
ring: "hsl(var(--ring) / <alpha-value>)",
chart: {
  "1": "hsl(var(--chart-1) / <alpha-value>)",
  "2": "hsl(var(--chart-2) / <alpha-value>)",
  "3": "hsl(var(--chart-3) / <alpha-value>)",
  "4": "hsl(var(--chart-4) / <alpha-value>)",
  "5": "hsl(var(--chart-5) / <alpha-value>)",
},
sidebar: {
  ring: "hsl(var(--sidebar-ring) / <alpha-value>)",
  DEFAULT: "hsl(var(--sidebar) / <alpha-value>)",
  foreground: "hsl(var(--sidebar-foreground) / <alpha-value>)",
  border: "hsl(var(--sidebar-border) / <alpha-value>)",
},
"sidebar-primary": {
  DEFAULT: "hsl(var(--sidebar-primary) / <alpha-value>)",
  foreground: "hsl(var(--sidebar-primary-foreground) / <alpha-value>)",
  border: "var(--sidebar-primary-border)",
},
"sidebar-accent": {
  DEFAULT: "hsl(var(--sidebar-accent) / <alpha-value>)",
  foreground: "hsl(var(--sidebar-accent-foreground) / <alpha-value>)",

```

```

    border: "var(--sidebar-accent-border)"
  },
  status: {
    online: "rgb(34 197 94)",
    away: "rgb(245 158 11)",
    busy: "rgb(239 68 68)",
    offline: "rgb(156 163 175)",
  },
},
fontFamily: {
  sans: ["var(--font-sans)"],
  serif: ["var(--font-serif)"],
  mono: ["var(--font-mono)"],
},
keyframes: {
  "accordion-down": {
    from: { height: "0" },
    to: { height: "var(--radix-accordion-content-height)" },
  },
  "accordion-up": {
    from: { height: "var(--radix-accordion-content-height)" },
    to: { height: "0" },
  },
},
animation: {
  "accordion-down": "accordion-down 0.2s ease-out",
  "accordion-up": "accordion-up 0.2s ease-out",
}

```

```

    },
    },
    },
    plugins: [require("tailwindcss-animate"), require("@tailwindcss/typography")],
  } satisfies Config;

```

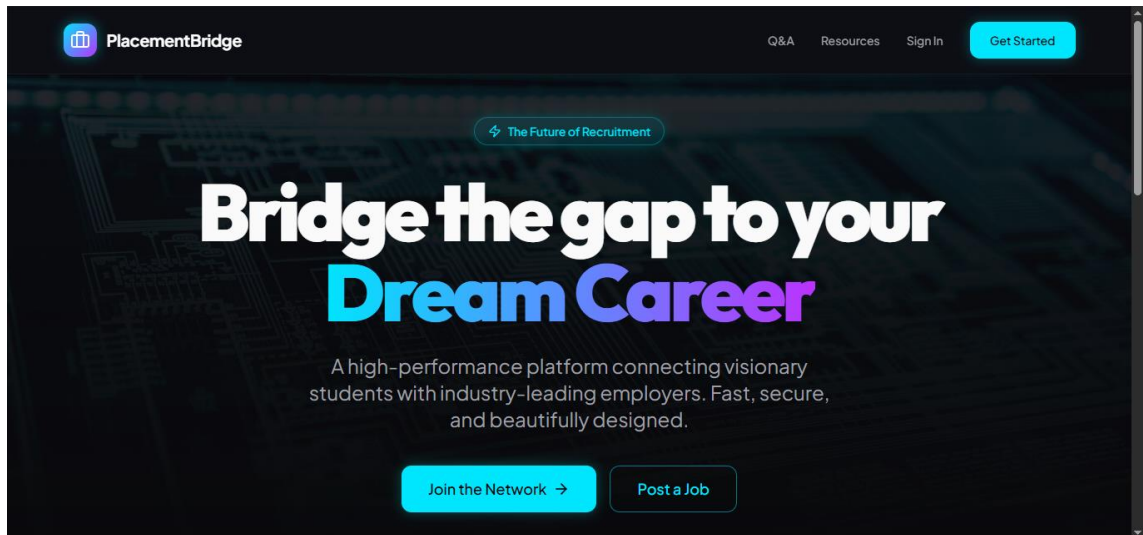


Figure 3: Home Page

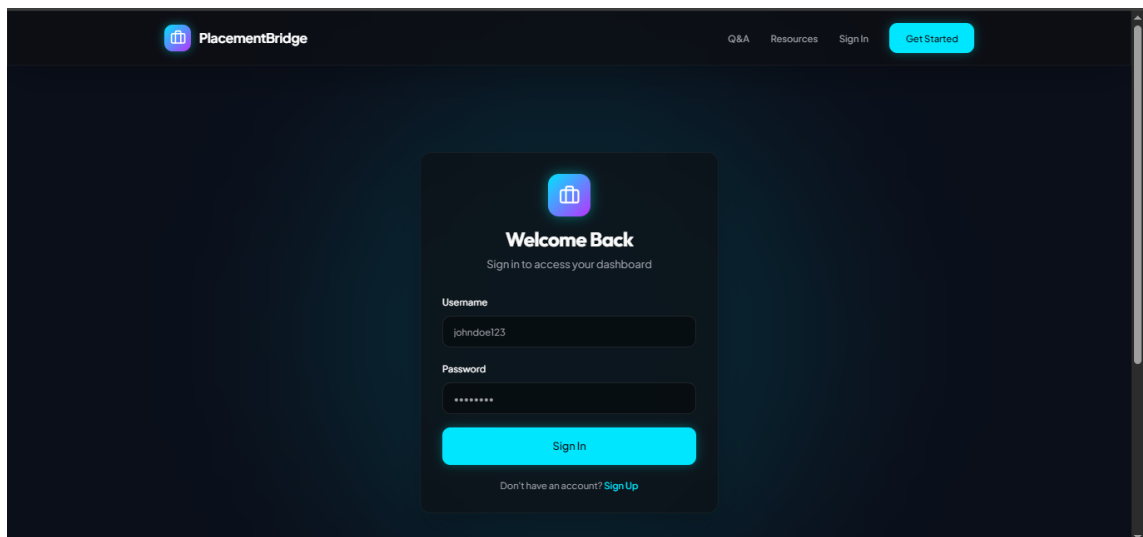


Figure 4: Login Page

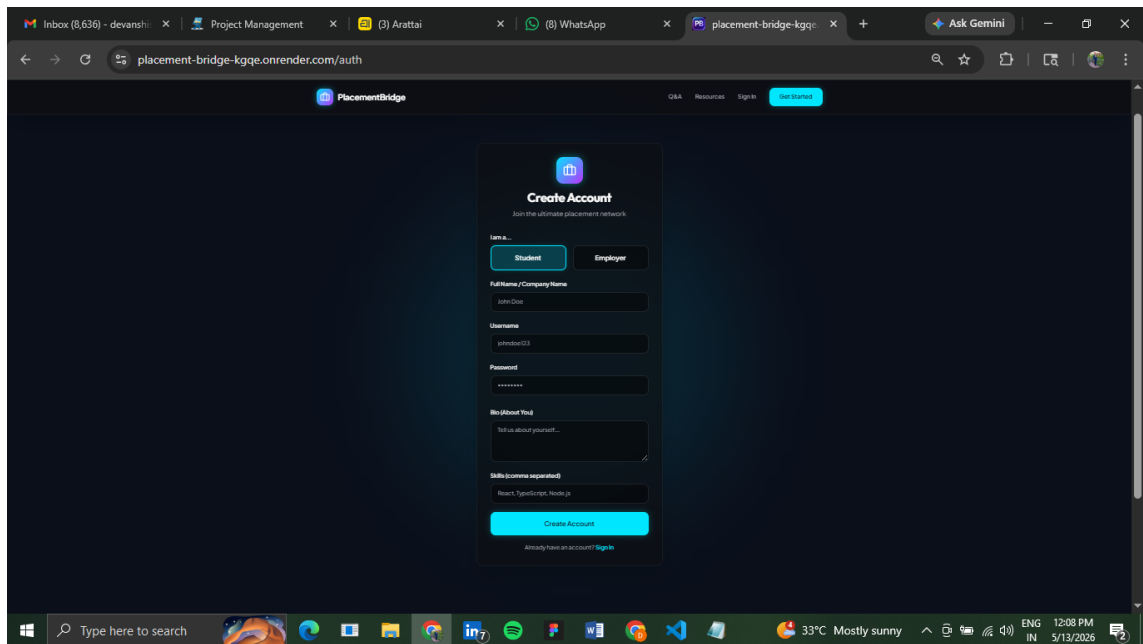


Figure 5: Sign Up Page

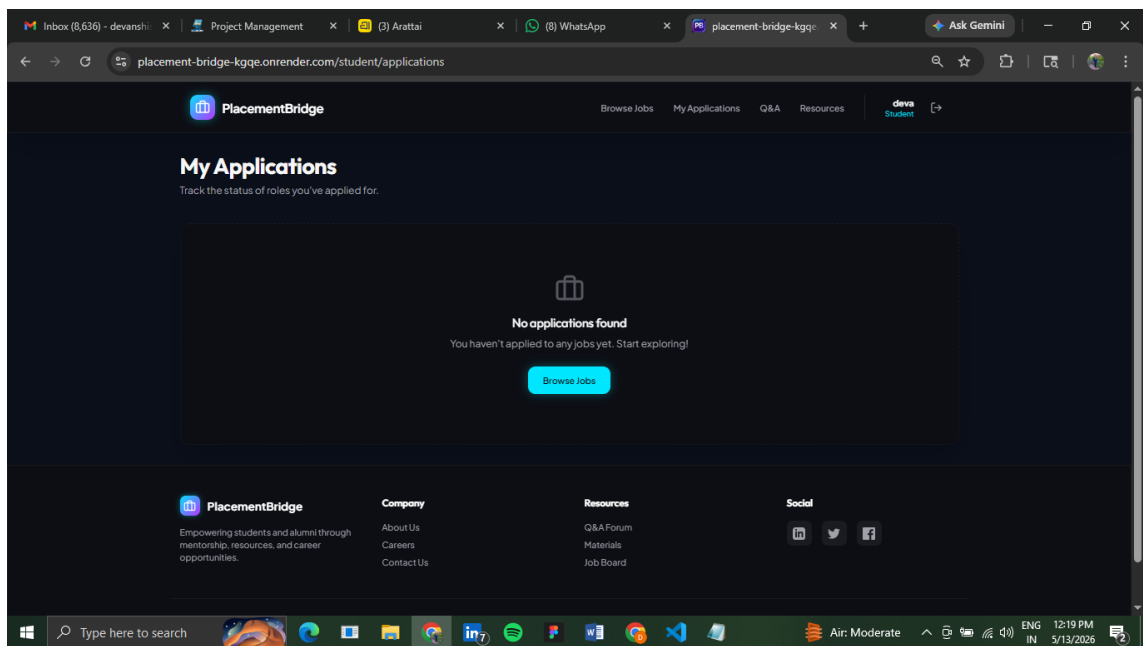


Figure 6: My Application Screen

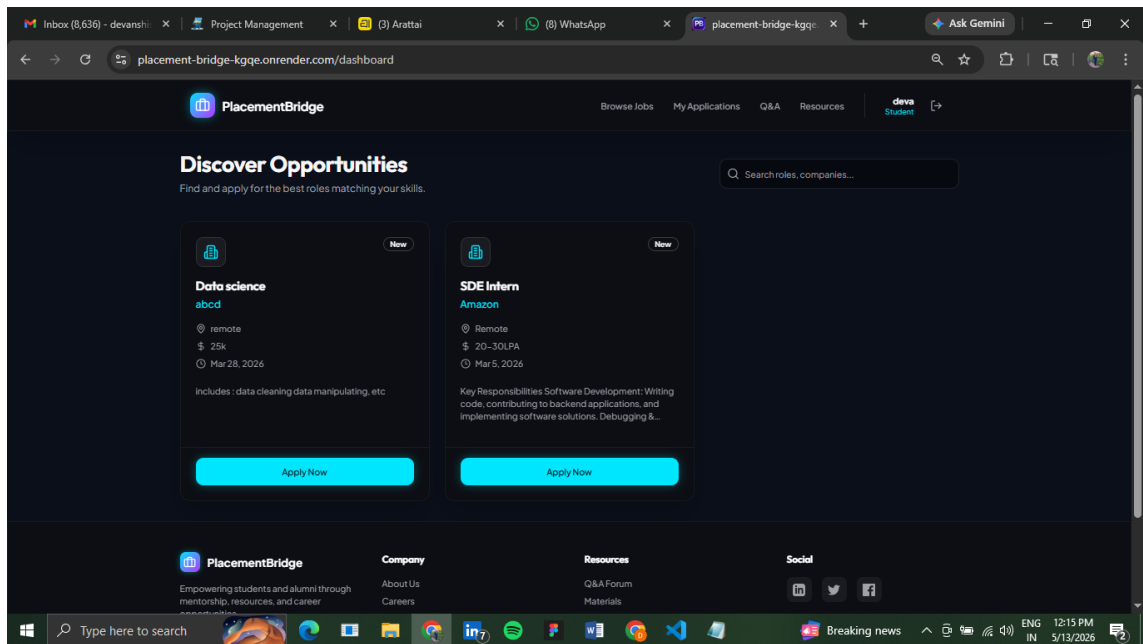


Figure 7: Discover Opportunity Page

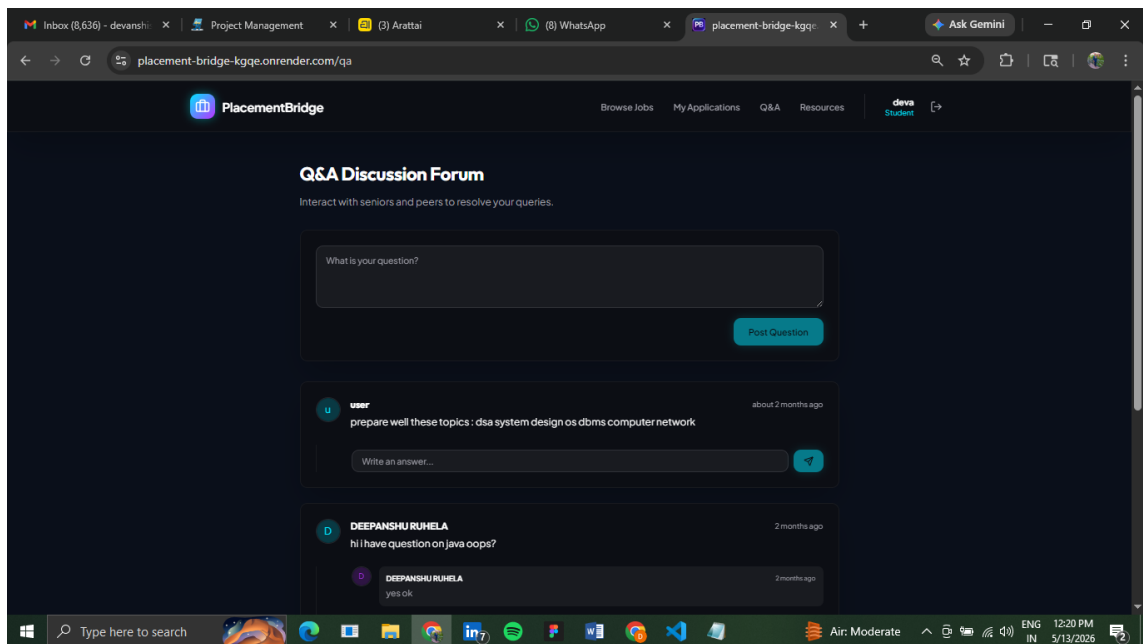


Figure 8 Discussion Forum

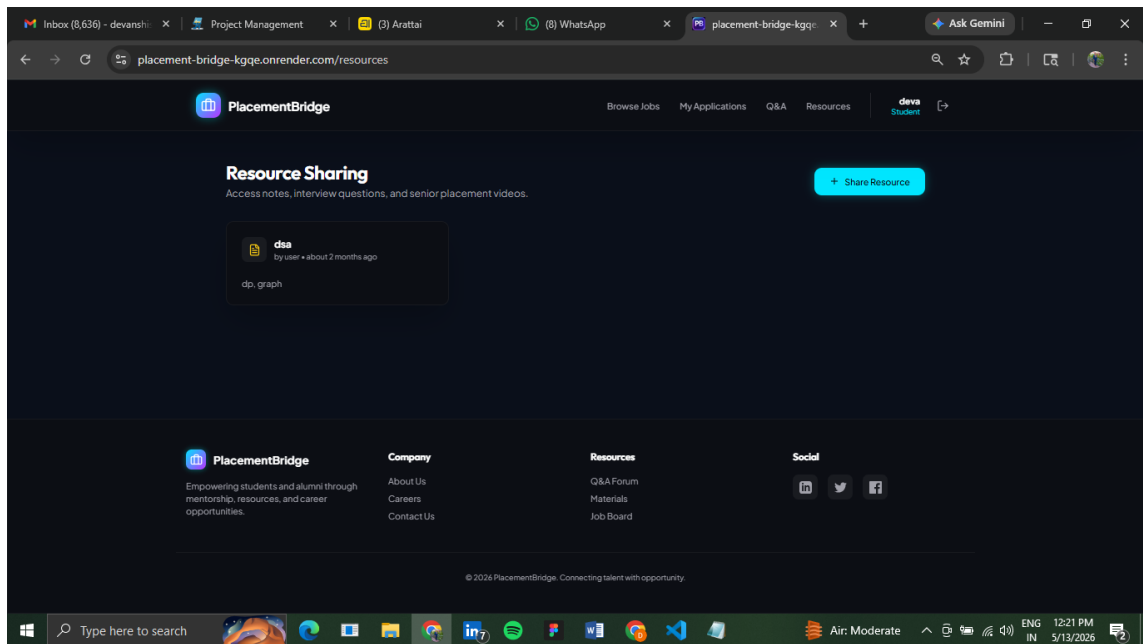


Figure 9 Resource Sharing Page

CHAPTER 7

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